Financial Constraints, R&D Investment, and Stock Returns

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Through the interaction between financial constraints and R&D, I study two asset-pricing puzzles: mixed evidence on the financial constraints–return relation and the positive R&D-return relation. Unlike capital investment, R&D is more inflexible. A financially constrained R&D-intensive firm is more likely to suspend/discontinue R&D projects. Therefore, R&D-intensive firms' risk increases with their financial constraints. Conversely, constrained firms' risk increases with their R&D intensity. I find a robust empirical relation between financial constraints and stock returns, primarily among R&D-intensive firms. Moreover, R&D predicts returns only among financially constrained firms. This evidence suggests that financial constraints potentially drive the positive R&D-return relation. (*JEL* G12, G32, O32)

Investment in research and development (R&D) is a key driver of long-term economic growth, and R&D-intensive firms constitute a large share of the stock market in the United States. Unlike capital expenditures, R&D investment is often much less flexible and often determined by science and/or regulation. If a firm cannot raise enough funds to conduct the required tests, it has to suspend the project. Suspension significantly reduces the firm's value because it prevents the resolution of the technical uncertainty and increases the likelihood that the firm will not be able to finish an R&D project before its competitors. Therefore, the impact of financial constraints is very severe for R&D-intensive firms.[1]

Despite its importance, R&D investment has attracted much less attention than capital expenditures. Moreover, classic models of R&D investment in

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[1] R&D-intensive firms are subject to more financial constraints as the literature documents that information asymmetry and agency problems are more severe for these firms (e.g., Hall 1992; Himmelberg and Petersen 1994; and Hall and Lerner 2010).

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finance (e.g., Berk, Green, and Naik 2004) usually assume, optimistically, that firms are always able to fund their projects in capital markets.

In this article, I focus on the more plausible case where R&D firms face funding constraints that may restrict their ability to finance new or ongoing projects at key stages. I characterize both the optimal behavior of R&D expenses and the expected equity returns to these firms, likely to be the most important component of their cost of capital. I show that expected returns are higher for financially constrained firms, which strengthens with the firm's R&D intensity. In other words, there is a strong interaction effect between financial constraints and R&D investment on expected returns.

These predictions are tested using a sample of R&D-reporting firms with the Fama-MacBeth (1973) cross-sectional regressions and portfolio sorts (Fama and French 1992, 1993). Firms with negative real sales growth are deleted to reduce the confounding effect of financial distress following the financial constraints literature (e.g., Kaplan and Zingales 1997; Lamont, Polk, and Saá-Requejo 2001; Whited and Wu 2006; Livdan, Sapriza, and Zhang 2009). Five proxies of financial constraints are used: the KZ index, the WW index, the SA index, age, and size measured by market capitalization.[2] Firms with a higher KZ index, higher WW index, higher SA index, younger age, and smaller size are more financially constrained than firms with lower KZ index, lower WW index, lower SA index, older age, and larger size. In addition, six measures of R&D intensity are used: R&D expenditure scaled by total assets, capital expenditure, sales, number of employees, and market equity, and R&D capital scaled by total assets. Following Chan, Lakonishok, and Sougiannis (2001), I compute R&D capital as the five-year cumulative R&D expenditures, assuming an annual depreciation rate of 20%.

The Fama-MacBeth regressions confirm the strong interaction effect between R&D intensity and financial constraints on firms' expected returns. For example, for the KZ index, the slope on the interaction term, $KZ * R\&D$, is positive and significant at the 1% or 5% level for all six measures of R&D intensity, whereas the slope on the KZ index is small and insignificant. This evidence implies that the KZ-return relation strengthens with the R&D intensity, and the R&D-return relation strengthens with the KZ index. If the KZ index is positively correlated with returns, this relation will manifest itself mostly among R&D-intensive firms. The results are similar for the other measures of financial constraints.

An alternative way of testing the model's predictions is by double sorting firms on financial constraints and R&D. As before, the prediction is that the constraints-return relation increases with R&D intensity and the R&D-return relation increases with the financial constraints. Such a portfolio-sorts-based test supports these predictions as well.

[2] The KZ index is from Kaplan and Zingales (1997) and Lamont, Polk, and Saá-Requejo (2001). The WW and SA indices are from Whited and Wu (2006) and Hadlock and Pierce (2010).

First, I find that the constraints-return relation among high R&D firms is positive and is much stronger than that among low R&D firms. For example, among firms with high R&D capital to assets (RDCA), the value-weighted monthly average return; the characteristic-adjusted return by size, book-to-market, and momentum; and the industry-adjusted return of the high-minus-low KZ portfolio are 0.60%, 0.47%, and 0.48%, with t-statistics of 2.00, 2.16, and 1.82, respectively. In contrast, among firms with low RDCA, the counterparts of these estimates are much lower and insignificant, 0.16%, 0.16%, and −0.01%, with t-statistics of 0.76, 0.97, and −0.05, respectively.[3]

Second, I find the positive R&D-return relation exists *only* among financially constrained firms. For example, among firms with a high SA index, the value-weighted monthly average return, characteristic-adjusted return, and industry-adjusted return of the high-minus-low RDME (R&D to market equity) portfolio are 1.30%, 1.17%, and 1.26%, respectively, and all are significant at the 1% level. In contrast, among firms with a low SA index, these estimates are 0.14%, 0.09%, and 0.19%, respectively, and none of them are significant.

The results are robust to using alternative measures of R&D intensity and financial constraints. For instance, the WW-return relation also increases with R&D intensity such as RDME. Among firms with high RDME, these return estimates for the high-minus-low WW portfolio are 0.82%, 0.87%, and 0.80%, with t-statistics of 2.28, 4.39, and 2.40, respectively. In contrast, among low RDME firms, the counterparts of these estimates are negative and insignificant, −0.25%, −0.20%, and −0.19%, with t-statistics of −0.87, −1.14, and −0.79, respectively. Conversely, the RDME-return relation exists only among high WW firms. In the high WW subsample, these estimates for the high-minus-low RDME portfolio are 1.22%, 1.15%, and 1.19%, respectively, and all are significant at the 1% level. In contrast, in the low WW subsample, these estimates are 0.15%, 0.08%, and 0.20%, respectively, and all are insignificant.

This article contributes to the literature on the relation between financial constraints and average stock returns (e.g., Lamont, Polk, and Saá-Requejo 2001; Whited and Wu 2006; Gomes, Yaron, and Zhang 2003, 2006; and Li, Livdan, and Zhang 2009). I argue that in previous studies, the mixed results for the constraints-return relation are attributed to the lack of controls for R&D investment. By modeling and empirically accounting for the significant impact of financial constraints on R&D investment due to the inflexibility of R&D, I examine this relation in R&D-reporting firms and document a robust positive constraints-return relation among high R&D firms. In a similar vein, Zhang (2005) uses an inflexibility argument to explain the value premium, and

3 The higher returns of the high-minus-low KZ portfolio formed among high RDCA firms indicate a stronger KZ-return relation rather than simply a larger variation in the KZ index. In fact, the spread in the KZ index among high RDCA firms is less than half of the spread among low RDCA firms: 20.87 versus 42.25. These results are consistent with the Fama-MacBeth regressions.

Gomes, Yaron, and Zhang (2006) and Livdan, Sapriza, and Zhang (2009) use it to explain why firms' risk should increase with their financial constraints status.

This article also relates to the literature on the relation between R&D intensity and stock returns (e.g., Chan, Martin, and Kensinger 1990; Chan, Lakonishok, and Sougiannis 2001; Chambers, Jennings, and Thompson 2002; Chu 2007; Lin 2007; Li and Liu 2010). I show that the positive R&D-return relation exists *only* among financially constrained firms. In addition, some measures of R&D intensity that cannot predict returns in the whole universe can do so among constrained firms. This evidence suggests that financial constraints potentially drive the positive R&D-return relation.

The article proceeds as follows. Section 1 describes the model and its main implications. Section 2 describes the data and discusses the empirical results. Section 3 concludes.

1. The Model

1.1 Overview

To illustrate the interaction effect of financial constraints and R&D, I add financial constraints to the R&D venture model of Berk, Green, and Naik (2004). In their model, the firm has no financial constraints and can always invest at the first-best level. However, insufficient funding is almost always a potential threat to an R&D venture's success and survival, as the venture often faces high financing costs due to information asymmetry and/or agency problems and huge demand for investment. Therefore, it is important to take into account financial constraints.

The firm works in continuous time and has a single multi-stage R&D project, which generates a stream of stochastic cash flows y_t after the firm successfully completes N discrete stages. The manager makes optimal investment decisions by maximizing the firm's intrinsic value subject to financial constraints. Similar to Berk, Green, and Naik (2004), the firm decides whether to suspend the project or to invest according to the requirements of science and/or regulation. The level of investment is not a choice variable. The firm will suspend the project if it cannot finance the required investment.

1.2 Valuation

The firm value at time t depends on the number of completed stages, n, and the future cash flow, $y(t)$, which follows a geometric Brownian motion:

$$dy(t) = \widehat{\mu} y(t)dt + \sigma y(t)d\widehat{w}(t).$$

Following many papers studying the cross-section of returns (e.g., Berk, Green, and Naik 1999, 2004; Carlson, Fisher, and Giammarino 2004, 2006; Zhang 2005; Livdan, Sapriza, and Zhang 2009), I adopt a partial equilibrium model with an exogenous pricing kernel. A partial equilibrium model provides the analytical tractability needed to focus on the dynamics of the relative risks of individual firms. The exogenous pricing kernel in this economy is given by the process

$$dm(t) = -rm(t)dt + \theta m(t)d\widehat{z}(t),$$

where r is the constant risk-free rate.[4] The market price of risk for $y(t)$ is computed as the covariance between the innovation of future cash flow and the innovation of the pricing kernel

$$\lambda = \sigma\theta\rho,$$

where ρ is the correlation between the two Brownian motion processes $\widehat{w}(t)$ and $\widehat{z}(t)$.

Under the risk-neutral measure, the cash flow process is given by

$$dy(t) = \mu y(t)dt + \sigma y(t)dw(t),$$

where $\mu = \widehat{\mu} - \lambda$, and $w(t)$ is a Brownian motion under the risk-neutral measure.[5]

After the firm completes the R&D project successfully, it receives a random stream of cash flows. Therefore, its value is given by the continuous-time version of the Gordon-Williams growth model, with a discount rate reflecting the risk of obsolescence and a risk-adjusted growth rate:[6]

$$V(y(t), N(t)) = \frac{y(t)}{r - \mu}.$$

For simplicity, I write $V(y(t), n(t))$ as $V(y, n)$ hereafter.

At any time t before the project is completed, the firm's value under the risk-neutral measure is the maximum of the following Bellman equation subject to the financial constraints

$$rV(y, n) = \max_{\upsilon \in \{0,1\}} -\upsilon x(n) + \frac{1}{dt}E_t[dV(y, n)] \tag{1}$$

$$s.t.\ p(n)\frac{1}{dt}E_t[dV(y, n)] \geq x(n), \tag{2}$$

where υ is the control variable, which equals 1 if the firm continues investing over the next instant and 0 otherwise. If the firm continues investing, it incurs

4 Since I only need the existence of a pricing kernel, I do not impose the complete markets assumption. However, even if I assume complete markets, no arbitrage and financial constraints can coexist since financial constraints in this model are in the firm's production set and are not limits to arbitrage.

5 I assume $\mu < r$ to ensure a finite firm value.

6 To reflect the risk of obsolescence, r can be set to a number higher than the risk-free rate.

an instantaneous cost $x(n)$. For simplicity, I sometimes subsume the number of completed stages, n. Notice that the level of investment is not a choice variable.

Equation (2) is the financial constraints. The firm can invest only if it raises enough funds to finance the required investment, $x(n)$. I assume that only a fraction (p) of the expected change in firm value ($\frac{1}{dt}E_t[dV]$) conditional on investing can be pledged as the "collateral" for external financing. Due to financial frictions caused by either hidden information, as in Greenwald, Stiglitz, and Weiss (1984) and Myers and Majluf (1984), or agency problems, as in Jensen and Meckling (1976), Grossman and Hart (1982), and Hart and Moore (1995), I assume that p lies between 0 and 1 and is a known function of n.[7] The technical complexity, high uncertainty, and long horizon associated with R&D and R&D-intensive firms' reluctance to fully reveal inside information for strategic reasons may aggravate the hidden information and agency problems. Therefore, ceteris paribus, the fraction p is lower for firms with more complex and uncertain technology. In the meantime, the amount of funds a firm can raise also depends on the expected change in firm value. The product of the fraction and the expected change in firm value jointly determine the upper bound of the firm's financing capacity. This specification is a parsimonious way to model the effect of financial frictions, as the focus here is not to identify the source of capital market imperfections, but rather to understand the effect of financial constraints on R&D investment and on firms' value and risk.

The Hamilton-Jacobi-Bellman (HJB) equation can be derived by applying Ito's lemma to the value function V and taking expectations:

$$rV(y,n) = \frac{1}{2}\sigma^2 y^2 \frac{\partial^2}{\partial y^2}V(y,n) + \mu y \frac{\partial}{\partial y}V(y,n) + \max_{\upsilon\in\{0,1\}} \upsilon\{\pi(n)[V(y,n+1) - V(y,n)] - x(n)\}. \tag{3}$$

The term in the curly brackets captures the cost-benefit analysis of the new R&D investment. The benefit of investing, $\pi(n)[V(y,n+1) - V(y,n)]$, is the expected jump in firm value if the firm advances to stage $n+1$ after the investment, where $\pi(n)$ is the success probability. A concrete example is a biotech firm's value typically jumping after it successfully finishes the Phase II trial and advances to Phase III.

In a perfect capital market, this analysis alone determines the investment decision. The firm will invest if future cash flow exceeds $y^*_{CB}(n)$, the threshold determined by the fundamentals. However, with financial frictions, the firm also needs to ensure that its financing capacity exceeds the required investment.

[7] Assuming no irrationality on the part of the investors, p cannot exceed 1. When p equals 1, from the Bellman equation, at $\upsilon = 1$, $\frac{1}{dt}E_t[dV] = rV + x$. Therefore, the financial constraints are always satisfied. If p equals 0, then the firm is never able to finance the R&D externally. In that case, no market exists for R&D projects at all. This example is the most extreme version of the lemons model in Akerlof (1970). Many factors can affect a firm's financing ability, such as information asymmetry, market liquidity, or even investors' tastes.

In other words, the cash flow also needs to exceed $y^*_{FC}(n)$, the threshold determined by financial constraints. Taking both into account, the firm will invest if the cash flow exceeds the threshold, $y^*(n)$, which is equal to $\max(y^*_{CB}(n), y^*_{FC}(n))$. For firms with low financing ability, $y^*_{FC}(n)$ tends to exceed $y^*_{CB}(n)$. Hence, financial constraints determine their investment decisions. For firms with high financing ability, the fundamentals play a more important role.

Following Berk, Green, and Naik (2004), I refer to the region where the firm invests as the "continuation" region and the region where the firm suspends the project as the "mothball" region. Appendix A details the valuation functions in both regions.

1.3 Risk premium

By standard arguments, the firm's risk premium (instantaneous expected rate of return in excess of the risk-free rate), R, at any stage is given by

$$\frac{V_y(y,n)y}{V(y,n)}\lambda. \tag{4}$$

After completion, the firm is equivalent to the underlying cash flow since no further investment decision is needed. Therefore, they have the same risk premium, λ. In the mothball region, the firm purely consists of an option to invest, which is riskier than the underlying asset due to implicit leverage. In the continuation region, the firm consists of the option to suspend, the discounted value of future cash flow, and the expected investment cost. Therefore, the firm is riskier than the underlying cash flow and less risky than the mothball region. The following propositions show how an R&D firm's risk premium varies with its financing ability $p(n)$ and investment level $x(n)$ in the continuation region.

1.3.1 Financing ability and risk premium. A firm is riskier if it needs to overcome a higher cash flow threshold in order to continue the project. For firms whose financial constraints determine their investment decisions, the cash flow threshold is inversely related to financing ability, p. Therefore, higher financing ability leads to lower risk premium. In other words, firms that are more constrained financially have higher expected returns. In addition, the relation intensifies with the required R&D investment, which is positively related to the threshold. However, for firms whose investment decisions are determined by the fundamentals, the relation between financing ability and returns is flat.

Proposition 1 formalizes this prediction for firms that have completed $N-1$ stages. The proof is in Appendix A.

Proposition 1. When $n = N - 1$ and the threshold $y^*_{FC}(n) > y^*_{CB}(n)$,

$$\frac{\partial R(n)}{\partial p(n)} < 0 \tag{5}$$

$$\frac{\partial^2 R(n)}{\partial p(n)\partial x(n)} < 0 \tag{6}$$

in the continuation region. If $y^*_{FC}(n) < y^*_{CB}(n)$, then $\frac{\partial R(n)}{\partial p(n)} = 0$.

To illustrate these effects for firms at other stages, I use numerical examples in which time is measured in years and the project involves five stages for completion. Given the required investments, firms differ in financing ability p. The drift (μ) and diffusion (σ) terms of the cash flow process are 3% and 40% per year, respectively. The risk-free rate incorporating the obsolescence risk is 17.54% per year, and the market price of risk for the cash flow process λ is 8% per year. After the firm completes the first stage, the success intensity $\pi(1)$ is 1, and it increases by 0.1 with each completed stage.[8] Therefore, after the firm completes the fourth stage, $\pi(4)$ becomes 1.3. The financing ability $p(n)$ ranges from 0.35 to 0.8 cross-sectionally. For simplicity, I make $p(n)$ constant over different stages for each firm. The required R&D investment increases by 3 with each completed stage and starts from 1 for low R&D firms and 10 for high R&D firms.

Figure 1 plots firms' risk premiums against their financing abilities p for different levels of R&D requirements and for different stages. In this example, future cash flow is so high that firms never need to mothball the project. When p is relatively low, the risk premium is negatively related to the financing ability for both the high and low investment levels. As p increases beyond the level above which financial constraints do not affect the firm's investment decision, this relation becomes flat.

Figure 1 also shows that this negative relation is stronger and lasts over a larger range of financing ability p for high R&D firms than for low R&D firms. In addition, the difference between the strengths of this relation becomes smaller as the firms complete more stages. This pattern is reasonable since the risk premium converges to the market price of risk for the cash flow λ as the firm gets closer to the completion of the project. The negative relation between the financing ability and the risk premium also weakens as firms mature. This weakness is due to the increase in firm value, which relaxes the financial constraints and reduces the possibility of suspension.

Instead of controlling for the number of stages a firm has completed, I illustrate the same intuition through the risk premium averaged over different

8 The assumption, $\pi(1) = 1$, corresponds to a 63.2% probability of completing at least one stage in a year. The results are robust to how the success intensity varies with each additional completed stage.

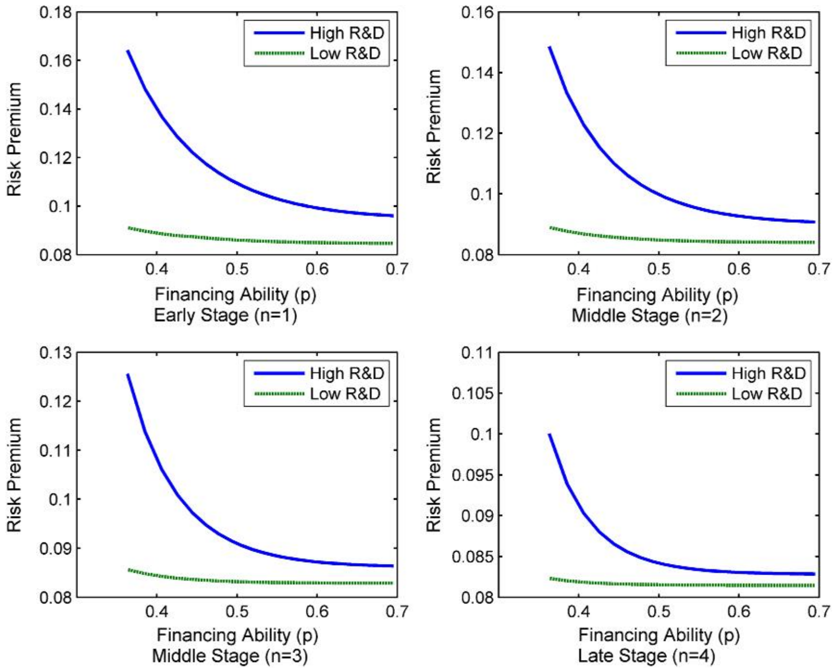

Figure 1
Financing ability and risk premium
The risk premiums (per annum) of projects that require five stages to complete are plotted as a function of their financing abilities (p) for different levels of R&D investment and over different stages. For example, the top left plot is for projects that have completed the first stage ($n = 1$), the top right for projects that have completed two stages ($n = 2$), and so forth. The risk premium is the instantaneous expected return minus the risk-free rate. The two lines correspond to different levels of R&D investment, which starts from 1 for low R&D firms and 10 for high R&D firms. The R&D investment increases by 3 with each additional completed stage. The volatility of future cash flow $\sigma = 0.4$ and the success intensity begin with $\pi(1) = 1$ and increase by 0.1 with each additional completed stage.

stages in Figure 2. The three levels of R&D (x) in the plot are 5, 10, and 15. The other parameter values are the same as before. The strength of the negative relation between financing ability (p) and risk premium obviously increases with the level of R&D.

This proposition implies that the positive constraints-returns relation should manifest itself most in R&D-intensive firms, which is consistent with the empirical findings.

1.3.2 R&D investment and risk premium. Similarly, a positive relation exists between R&D investment and the risk premium in the continuation region for firms with $y^*_{FC}(n) > y^*_{CB}(n)$. This relation is stronger among firms with lower financing abilities. Proposition 2 formalizes this prediction for $n = N - 1$, and Appendix A shows the proof.

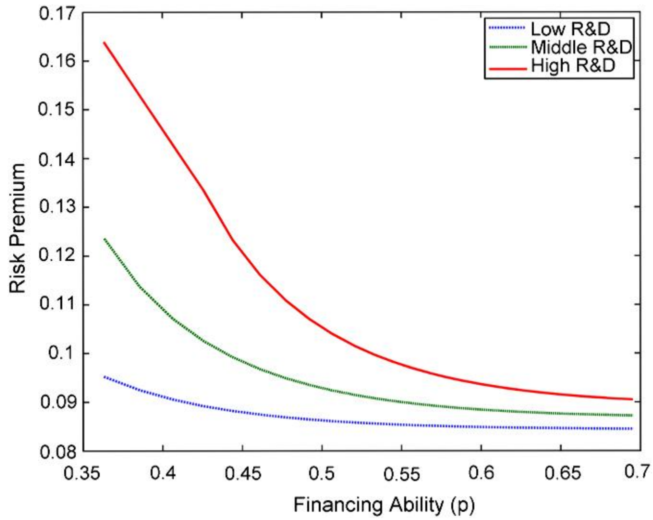

Figure 2
Financing ability and time-averaged risk premium
This figure plots the time-averaged risk premium against the financing ability (p) for firms with different levels of R&D investment (x). The project takes five stages to complete. The risk premium is averaged over the five stages. The three levels of R&D (x) start from 5, 10, and 15, respectively, and increase by 3 with each additional completed stage. The volatility of future cash flow $\pi = 0.4$ and the success intensity begin with $\pi(1) = 1$ and increase by 0.1 with each additional completed stage.

Proposition 2. When $n = N - 1$ and the threshold $y^*_{FC}(n) > y^*_{CB}(n)$,

$$\frac{\partial R(n)}{\partial x(n)} > 0 \tag{7}$$

$$\frac{\partial^2 R(n)}{\partial x(n) \partial p(n)} < 0 \tag{8}$$

in the continuation region.

The intuition is similar to Proposition 1 because the effect of a high required investment x on a firm's investment decision is similar to that of a low p. Ceteris paribus, a firm with a higher required investment is more likely to mothball the project due to insufficient funds in the event of an adverse shock to future cash flow. Therefore, its investment decisions and value are more sensitive to the systematic risk the cash flow carries. Furthermore, this relation is stronger for firms with lower financing ability p since a decrease in p intensifies the sensitivity to future cash flow. However, the theory does not necessarily predict a monotonic relation between R&D and returns for firms with $y^*_{FC}(n) < y^*_{CB}(n)$.

Numerical examples are used to illustrate these effects at other stages. Figure 3 plots firms' risk premiums against their investment requirements x for

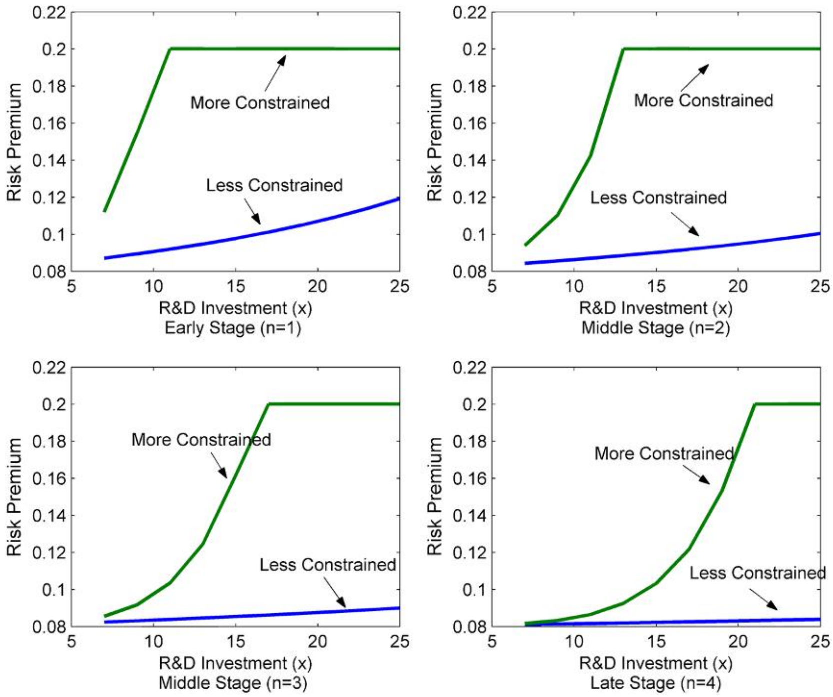

Figure 3
R&D investment and risk premium
The risk premiums (per annum) of projects that require five stages to complete are plotted as a function of their R&D investment requirement (x) for different financing abilities (p) and over different stages. For example, the top left plot is for projects that have completed the first stage ($n = 1$), the top right for projects that have completed two stages ($n = 2$), and so forth. The risk premium is the instantaneous expected return minus the risk-free rate. The two lines correspond to different levels of external financing abilities (p): 0.3 for more constrained firms and 0.7 for less constrained firms. The volatility of future cash flow $\sigma = 0.4$ and the success intensity begin with $\pi(1) = 1$ and increase by 0.1 with each additional completed stage.

different levels of financing ability p and for different stages.[9] The horizontal parts in the plot correspond to the mothball regions. We see that R&D is positively related to risk premiums in the continuation regions. Furthermore, this relation is stronger for more constrained firms. Similarly, as firms get closer to completion, this positive relation weakens because the threat of suspension due to insufficient funds decreases as the firm's value increases.

Figure 4 illustrates the same intuition through the averaged risk premium without controlling the number of completed stages. The strength of the positive relation between R&D (x) and the risk premium clearly decreases with the level of financing ability (p).[10]

9 The required investment x ranges from 7 to 25 cross-sectionally. The financing ability p is constant over the stages. The low p equals 0.3, and the high p equals 0.7. All other parameters are the same as before.

10 In Figure 4, the three levels of financing ability (p) in the plot are 0.3, 0.5, and 0.7, respectively. The success density, π, starts from 1 and increases by 0.1 with each additional completed stage. These numerical results are robust to many different values for the key parameters, such as π and σ.

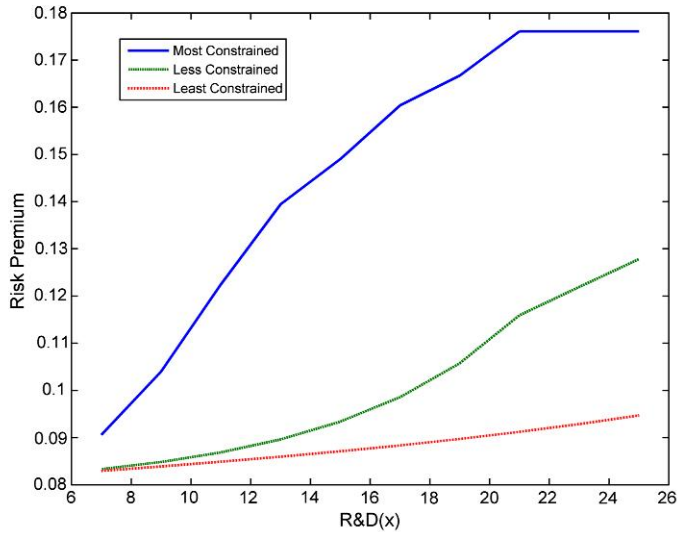

Figure 4
R&D investment and time-averaged risk premium
This figure plots the time-averaged risk premium against the R&D investment (x) for firms with different levels of financing ability (p). The project takes five stages to complete. The risk premium is averaged over the five stages. The three levels of financing ability (p) are 0.3, 0.5, and 0.7. The volatility of future cash flow $\sigma = 0.4$ and the success intensity begin with $\pi(1) = 1$ and increase by 0.1 with each additional completed stage.

Although this model describes single-project R&D ventures (i.e., small R&D-intensive firms), the insight and implications apply to R&D-intensive firms with multiple projects and internal cash flows. For those firms, I can treat all the incomplete R&D projects as one "composite" project. The investment requirement for this project is the gap between the combined investment requirement and the internal cash flows. As long as the internal cash flows do not always meet the investment requirement and the firm value depends heavily on the progress of this "composite" project, the firm will be subject to the risk of project suspension, which will reduce the firm's value drastically. Therefore, the empirical implications go beyond single-project R&D ventures.

2. Empirical Analysis

In this section, I test the model's implications through the Fama-MacBeth (1973) regressions and portfolio sorts. The model predicts that the constraints-return relation strengthens with firms' R&D intensity, and the R&D-return relation strengthens with firms' financial constraints status.

2.1 Data and measures of financial constraints and R&D intensity

Accounting data are obtained from Compustat, and stock returns data from the Center for Research in Security Prices (CRSP). All domestic common shares trading on NYSE, AMEX, and NASDAQ with available accounting and returns data are included. The sample is from 1975 to 2007 since the accounting treatment of R&D expense reporting was standardized in 1975 (Financial Accounting Standards Board Statement No. 2). I use annual R&D expenditure since quarterly R&D data are unavailable until 1989. The sample only includes firm-year combinations with non-missing R&D expenditure. To reduce the confounding effect of financial distress, I also delete observations with negative real sales growth following the literature on financial constraints (e.g., Kaplan and Zingales 1997; Lamont, Polk, and Saá-Requejo 2001; Whited and Wu 2006; and Livdan, Sapriza, and Zhang 2009). These two restrictions delete 3,287 firms each year on average, as reported in Table 1. The average and median market capitalization of the deleted firms are \$940 million and \$100 million, respectively.

Compared with deleted firms, the R&D-reporting firms with positive real sales growth are much bigger, with an average and median size of \$1,740 million and \$180 million, respectively. Although the sample on average includes 1,333 firms each year, it covers 43% of the total U.S. market capitalization. In general, high R&D firms are smaller than low R&D firms in median size, but similar in average size.

Table 1
Summary statistics

	No. of Firms	Average *ME* (\$bn)	Median *ME* (\$bn)	Total *ME* (\$bn)	*ME* (%)
Non-R&D firms and R&D firms with $RSGRO \leq 0$	3287	0.94	0.10	3083	57
All R&D firms with $RSGRO > 0$	1333	1.74	0.18	2325	43
Low *RDA*	650	1.79	0.24	1166	22
High *RDA*	651	1.74	0.15	1130	21
Low *RDCA*	653	1.75	0.24	1142	21
High *RDCA*	654	1.78	0.15	1164	22
Low *RDCAP*	641	2.00	0.26	1285	24
High *RDCAP*	642	1.54	0.14	990	18
Low *RDS*	650	1.79	0.23	1161	21
High *RDS*	651	1.74	0.16	1135	21
Low *RDE*	640	1.51	0.22	966	18
High *RDE*	641	2.04	0.17	1310	24
Low *RDME*	645	2.33	0.27	1501	28
High *RDME*	645	1.22	0.13	789	15

At the end of June of year t, I sort R&D-reporting firms with positive real sales growth (*RSGRO*) into low and high R&D groups according to their R&D intensities in the fiscal year ending in year $t-1$. The portfolios are reformed every year. *RDA*, *RDCAP*, *RDS*, *RDE*, and *RDME* are R&D expenditure scaled by assets, capital expenditure, sales, number of employees, and year-end market equity. *RDCA* is R&D capital scaled by assets, where R&D capital is the weighted sum of a firm's R&D expenditure over the past five years, assuming an annual amortization rate of 20%. *ME* is year-end market equity. I report the time-series mean of cross-sectional average number of firms, the mean and median market capitalization (in billions), total market capitalization, and the market capitalization percentage of all R&D firms and of low and high R&D firms. For comparison, I also report these summary statistics of non-R&D reporting firms and R&D-reporting firms with negative real sales growth in the top row. The sample period is from 1975 to 2007.

Five measures of financial constraints are used: the KZ index, the WW index, the SA index, firm age, and size. The KZ index is a linear combination of the following variables with signs in parentheses: debt to total capital (+), dividends to capital (−), cash holdings to capital (−), cash flow to capital (−), and Tobin's Q (+). The WW index is a linear combination of cash flow to total assets (−), sales growth (−), long-term debt to total assets (+), log of total assets (−), dividend policy indicator (−), and the firm's three-digit industry sales growth (+). The SA index is a combination of asset size and firm age. By construction, these indices are higher for more financially constrained firms. Appendix B provides further details on how to construct them. Age is the number of years a firm is listed with a non-missing stock price on Compustat.[11] Size is the market capitalization of equity and is a popular measure of financial constraints (e.g., Gertler and Gilchrist 1994; Hubbard 1998; Campello and Chen 2005; Livdan, Sapriza, and Zhang 2009; Hadlock and Pierce 2010). Young and smaller firms are more financially constrained.

Six measures of R&D intensity are used: R&D expenditure scaled by total assets (*RDA*), capital expenditure (*RDCAP*), sales (*RDS*), number of employees (*RDE*), and market equity (*RDME*), and R&D capital scaled by total assets (*RDCA*).[12] Following Chan, Lakonishok, and Sougiannis (2001), I compute R&D capital as the five-year cumulative R&D expenditures assuming an annual depreciation rate of 20%. Specifically, R&D capital for firm i in year t, RDC_{it}, is a weighted average of annual R&D expenditures over the last five years:

$$RDC_{it} = RD_{it} + 0.8 * RD_{it-1} + 0.6 * RD_{it-2} + 0.4 * RD_{it-3} + 0.2 * RD_{it-4.}$$

In unreported results, I find the Spearman's rank correlations among the six R&D measures range from 0.70 to 0.96. Similarly, the five measures of financial constraints are also highly correlated with each other except the KZ index. For example, the rank correlation between size and the WW index, the SA index, and age are −0.82, −0.74, and 0.39, respectively. However, the correlation between size and the KZ index is only −0.22.

2.2 Fama-MacBeth regressions

Before moving on to the new results, I first replicate the slightly negative KZ-return relation from Lamont, Polk, and Saá-Requejo (2001; LPS) by estimating monthly Fama-MacBeth (FM) cross-sectional regressions in the following form:

$$\begin{aligned} R = \alpha + \gamma_1 * Constraints + \gamma_2 * \ln(ME) + \gamma_3 * \ln(BE/ME) \\ + \gamma_4 * Momentum + \gamma_5 * ROA, \end{aligned} \tag{9}$$

[11] Following Hadlock and Pierce (2010), I winsorize age at 37 years.

[12] Previous studies have used many of these measures (e.g., Lev and Sougiannis 1996, 1999; Kothari, Laguerre, and Leone 2002).

where R is individual stocks' returns from July of year t to June of year $t+1$, $Constraints$ is measured by the KZ index in the fiscal year ending in year $t-1$, ln(*ME*) is the natural log of market capitalization at the end of June of year t, ln(*BE/ME*) is the natural log of the ratio of book equity to market equity for the fiscal year ending in year $t-1$, *Momentum* is the prior six-month returns (with a one-month gap between the holding period and the current month), and *ROA* is income before extraordinary items for the fiscal year ending in year $t-1$ divided by total assets for the fiscal year ending in year $t-2$.

Consistent with LPS, the average slope on the KZ index is negative and insignificant in the LPS sample and the extended sample used in this article, as shown in Panels A and B of Table 2. The LPS sample includes manufacturing firms with positive real sales growth from 1968 to 1997, and the extended sample includes all firms with positive real sales growth from 1975 to 2007.

Table 2
Comparison of Fama-MacBeth regression results in different samples

Panel A. The KZ-return relation in LPS (2001) sample

KZ	ln(*BE/ME*)	ln(*ME*)	*Momentum*	*ROA*
–0.19	0.44	–0.31	0.88	0.75
(–1.21)	(5.79)	(–0.95)	(3.29)	(1.25)

Panel B. The KZ-return relation in the extended sample

KZ	ln(*BE/ME*)	ln(*ME*)	*Momentum*	*ROA*
–0.29	0.36	−1.06	0.70	0.22
(–1.43)	(5.41)	(–3.02)	(2.88)	(0.47)

Panel C. The R&D-return relation in Chan, Lakonishok, and Sougiannis (2001) sample

ln(*BE/ME*)	ln(*ME*)	*Momentum*	*RDME*	*ROA*	*RDS*
0.29	−1.12	0.39	1.06	0.78	
(3.85)	(–3.02)	(1.79)	(5.30)	(1.83)	
0.34	−1.23	0.40		0.69	0.80
(5.49)	(–3.28)	(1.81)		(1.62)	(3.31)

Panel D. The R&D-return relation in the extended sample

ln(*BE/ME*)	ln(*ME*)	*Momentum*	*RDME*	*ROA*	*RDS*
0.31	−0.93	0.67	1.27	0.48	
(4.24)	(–2.67)	(2.75)	(4.82)	(1.17)	
0.41	−1.03	0.65		0.38	0.96
(6.83)	(–2.91)	(2.74)		(0.93)	(2.85)

I report the time-series average slopes and their t-statistics (in parentheses) from Fama-MacBeth cross-sectional regressions within different samples. The Lamont, Polk, and Saá-Requejo (LPS, 2001) sample in Panel A includes manufacturing firms with positive real sales growth from 1968 to 1997. The extended sample in Panels B and D includes all firms with positive real sales growth from 1975 to 2007. The Chan, Lakonishok, and Sougiannis (2001) sample in Panel C includes all firms from 1975 to 1995. For each month from July of year t to June of year $t+1$, I run cross-sectional regressions of monthly percent returns on different variables: *KZ* is the Kaplan and Zingales (1997) index of financial constraints for the fiscal year ending in year $t-1$; ln(*BE/ME*) is the log book equity for the fiscal year ending in year $t-1$ minus the log market equity at the end of December of year $t-1$; ln(*ME*) is the log market capitalization at the end of June of year t; *Momentum* is the prior six-month returns (with a one-month gap between the holding period and the current month); *ROA* is the net income scaled by total assets for the fiscal year ending in year $t-1$; *RDME* and *RDS* are R&D expenditure scaled by year-end market equity and sales, respectively, for the fiscal year ending in year $t-1$. *KZ*, *RDME*, *RDS*, and ln(*ME*) are demeaned percentiles. The other control variables are winsorized at the top and bottom 1%.

I then replicate the positive R&D-return relation from Chan, Lakonishok, and Sougiannis (2001; CLS) by estimating the FM regressions in the following form:

$$R = \alpha + \gamma_1 * R\&D + \gamma_2 * \ln(ME) + \gamma_3 * \ln(BE/ME) + \gamma_4 * Momentum + \gamma_5 * ROA, \tag{10}$$

where *R&D* is measured by *RDME* or *RDS* in the fiscal year ending in year $t-1$, and all the other variables are measured in the same way as in (9). I confirm the positive R&D-return relation in both the CLS sample, which includes all firms with available data from 1975 to 1995, and the extended sample defined previously. Panels C and D of Table 2 show that the slope on *RDME* and *RDS* is significantly positive in both samples.

I now move to test my model, which predicts that the constraints-return relation strengthens with firms' R&D intensity, and the R&D-return relation strengthens with firms' financial constraints status. In other words, the model implies a significantly positive interaction effect between R&D and financial constraints. To test this prediction, I estimate the FM cross-sectional regressions in the following form for each month from July of year t to June of year $t+1$ in the extended sample:

$$R = \alpha + \gamma_1 * Constraints + \gamma_2 * R\&D + \gamma_3 * Constraints * R\&D + \gamma_4 * ROA + \gamma_5 * \ln(ME) + \gamma_6 * \ln(BE/ME) + \gamma_7 * Momentum, \tag{11}$$

where all the variables are measured in the same way as in (9) and (10). Notice that (11) nests both (9) and (10) and includes the extra interaction term, $Constraints * R\&D$, which captures the interaction effect between financial constraints and R&D.

The model predicts a significantly positive slope on the interaction term. Table 3 confirms this prediction for the KZ index. Panel A shows that the average slope on the interaction term, $KZ * R\&D$, is positive and statistically significant at the 1% or 5% level for different measures of R&D intensity. In unreported results, I find that the significant interaction effect is robust to alternative measures of financial constraints. Consistent with previous studies, the slope on the KZ index is insignificant, and the slope on R&D is positive and significant at the 1% level. The slopes on ln(*ME*), ln(*BE/ME*), and *Momentum* are negative, positive, and positive, respectively, and are significant at the 1% level. The slope on *ROA* is positive and marginally significant.

Panel B in Table 3 includes an additional term, $KZ * R\&D * ln(ME)$, to check the effect of size on the interaction effect between the KZ index and R&D intensity. Panel B shows that the slope on $KZ * R\&D$ remains significantly positive, and the slope on $KZ * R\&D * ln(ME)$ is negative but insignificant. The pattern is similar for the other measures of financial constraints in unreported results. In particular, the slope on the interaction

Table 3
Slopes from Fama-MacBeth (1973) cross-sectional regressions of monthly percent returns on the KZ index, R&D intensity, KZ *R&D, and other control variables

Panel A: Without the interaction term *KZ*R&D*ln(ME)*

	KZ	*R&D*	*KZ* R&D*	*KZ* R&D* ln(ME)*	*ROA*	*ln(ME)*	*ln(BE/ME)*	*Momentum*
RDME	0.02	1.26	1.17		0.70	−0.93	0.32	0.65
	(0.09)	(5.15)	(2.71)		(1.62)	(−2.68)	(4.60)	(2.72)
RDS	0.07	1.01	0.95		0.77	−1.03	0.42	0.64
	(0.46)	(3.06)	(2.40)		(1.83)	(−2.91)	(7.19)	(2.72)
RDE	0.10	1.08	1.06		0.69	−1.10	0.41	0.65
	(0.61)	(3.40)	(2.67)		(1.61)	(−3.10)	(6.87)	(2.77)
RDCA	0.10	1.19	0.78		0.69	−0.89	0.42	0.65
	(0.60)	(4.30)	(2.04)		(1.64)	(−2.58)	(7.17)	(2.77)
RDCAP	0.08	1.02	1.15		0.73	0.94	0.42	0.68
	(0.52)	(3.89)	(2.83)		(1.69)	(−2.68)	(6.90)	(2.84)
RDA	0.12	1.21	0.80		0.77	−0.96	0.44	0.64
	(0.78)	(4.02)	(1.96)		(1.83)	(−2.76)	(7.42)	(2.71)

Panel B: With the interaction term *KZ*R&D*ln(ME)*

	KZ	*R&D*	*KZ* R&D*	*KZ* R&D* ln(ME)*	*ROA*	*ln(ME)*	*ln(BE/ME)*	*Momentum*
RDME	0.00	1.22	1.03	−1.81	0.74	−0.94	0.32	0.66
	(−0.02)	(5.01)	(2.57)	(−1.15)	(1.69)	(−2.69)	(4.55)	(2.73)
RDS	0.07	0.96	0.90	−1.40	0.78	−1.03	0.42	0.65
	(0.46)	(2.93)	(2.34)	(−1.81)	(1.87)	(−2.91)	(7.20)	(2.73)
RDE	0.09	1.04	1.04	−1.56	0.70	−1.12	0.41	0.65
	(0.61)	(3.26)	(2.62)	(−0.89)	(1.64)	(−3.11)	(6.89)	(2.77)
RDCA	0.09	1.13	0.70	−2.90	0.72	−0.93	0.43	0.66
	(0.58)	(4.08)	(1.92)	(−1.90)	(1.70)	(−2.68)	(7.20)	(2.78)
RDCAP	0.08	0.99	1.06	−1.21	0.74	−0.95	0.42	0.69
	(0.47)	(3.77)	(2.74)	(−0.74)	(1.70)	(−2.71)	(6.88)	(2.85)
RDA	0.12	1.15	0.73	−2.59	0.79	−1.00	0.44	0.65
	(0.75)	(3.81)	(1.86)	(−1.57)	(1.87)	(−2.83)	(7.43)	(2.72)

In Panel A for each month from July of year t to June of year t+1 I run cross-sectional regressions of monthly percent returns on *KZ*, *R&D*, and *KZ*R&D* measured in the fiscal year ending in year $t-1$, and other control variables: *ROA* is the net income scaled by total assets for the fiscal year ending in year $t-1$, *ln(ME)* is the natural log of market capitalization at the end of June of year t, *ln(BE/ME)* is the natural log of the ratio of book equity to market equity for the fiscal year ending in year $t-1$, and *Momentum* is the prior six-month returns (with a one-month gap between the holding period and the current month). The first column shows measures of R&D intensity used in the regressions. *RDA*, *RDCAP*, *RDS*, *RDE*, and *RDME* are defined as in Table 1. *RDCA* is R&D capital scaled by assets, where R&D capital is the weighted sum of a firm's R&D expenditure over the past five years, assuming an annual amortization rate of 20%. In Panel B, an additional interaction term is included: *KZ*R&D*ln(ME)*. *KZ*, *RDME*, *RDS*, and *ln(ME)* are demeaned percentiles. The other control variables are winsorized at the top and bottom 1%. I report the slopes and their Fama-MacBeth t-statistics (in parentheses). The sample is from 1975 to 2007 and includes R&D-reporting firms with positive real sales growth only.

term, $Constraints * R\&D$, remains significantly positive, and the slope on the three-way interaction term, $Constraints * R\&D * ln(ME)$, is either insignificant or significantly positive. These results suggest that extremely small firms are unlikely to drive the strong interaction between financial constraints and R&D. This is also consistent with the triple sorts presented later, which shows that the positive KZ-return relation is more common in large high R&D firms.

2.3 Portfolio analysis

At the end of June of year t, I sort firms independently into two R&D groups and three financial constraints groups. The intersection forms six R&D-constraints portfolios. All ranking variables are measured in the fiscal year ending in year $t-1$ except size, which is measured at the end of June of year t. I also form a zero-investment portfolio that goes long on the constrained portfolio and short on the unconstrained portfolio within each R&D group and a zero-investment portfolio that goes long on the high R&D portfolio and short on the low R&D portfolio within each constraints group. I hold these portfolios over the next 12 months and rebalance them each year. Value-weighted monthly average returns, industry-adjusted returns, and returns adjusted by size, book-to-market, and momentum are computed for each portfolio. The industry-adjusted returns are based on the difference between individual firms' returns and returns of matching industry portfolios based on two-digit SIC codes. Following Daniel, Grinblatt, Titman, and Wermers (DGTW 1997) and Wermers (2004), the characteristic-adjusted returns are based on the difference between individual firms' returns and the DGTW benchmark portfolio returns.[13]

2.3.1 Variation of the constraints-return relation with R&D. Table 4 shows that the constraints-return relation strengthens with R&D intensity. Panel A reports the results for the KZ index. The returns of the high-minus-low KZ portfolios are in general significantly positive in the high R&D group but insignificant in the low R&D group. For example, when R&D intensity is measured by R&D to sales (RDS), the value-weighted monthly average return, the characteristic-adjusted return, and the industry-adjusted return of the high-minus-low KZ portfolio are 0.57%, 0.44%, and 0.42%, with t-statistics of 1.99, 2.00, and 1.64, respectively, in the high RDS group. In contrast, these estimates in the low RDS group are only 0.18%, 0.17%, and 0.06%, with t-statistics of 0.86, 1.01, and 0.42, respectively.

Panel B of Table 4 reports the results for the WW index. The returns of the high-minus-low WW portfolios formed in the high R&D group are much larger than those formed in the low R&D group. Furthermore, the characteristic-adjusted returns of the hedge portfolios are statistically significant at the 1% level in the high R&D group, but insignificant in the low R&D group for all measures of R&D intensity. Specifically, the monthly value-weighted characteristic-adjusted returns of the high-minus-low WW portfolio are 0.54%, 0.54%, 0.51%, 0.87%, 0.52%, and 0.52% in the high RDCA, high RDS, high RDCAP, high RDME, high RDE, and high RDA groups, respectively. All are significant at the 1% level. In contrast, the counterparts of these estimates are only 0.11%, 0.08%, 0.16%, −0.20%, 0.16%, and 0.09% in the low RDCA, low

[13] The DGTW benchmarks are available via http://www.smith.umd.edu/faculty/rwermers/ftpsite/Dgtw/coverpage.htm.

Table 4
The returns of the constrained-minus-unconstrained portfolios across subsamples split by R&D

	Panel A: KZ Index			Panel B: WW Index			Panel C: SA Index			Panel D: Size			Panel E: Age		
	Raw return	Charact.-adjusted return	Industry-adjusted return	Raw return	Charact.-adjusted return	Industry-adjusted return	Raw return	Charact.-adjusted return	Industry-adjusted return	Raw return	Charact.-adjusted return	Industry-adjusted return	Raw return	Charact.-adjusted return	Industry-adjusted return
Low *RDCA*,	0.16	0.16	−0.01	0.11	0.11	0.21	−0.17	−0.03	−0.04	0.72	0.42	0.79	−0.36	−0.38	−0.12
const.-unconst	(0.76)	(0.97)	(−0.05)	(0.37)	(0.59)	(0.89)	(−0.54)	(−0.17)	(−0.17)	(2.70)	(2.85)	(3.41)	(−1.32)	(−2.16)	(0.64)
High *RDCA*,	0.60	0.47	0.48	0.53	0.54	0.47	0.57	0.48	0.47	1.45	1.22	1.33	0.72	0.28	0.63
const.-unconst.	(2.00)	(2.16)	1.82	(1.45)	(2.73)	(1.36)	(1.53)	(2.49)	(1.42)	(4.13)	(6.78)	(4.06)	(1.87)	(1.07)	(1.93)
Low *RDS*,	0.18	0.17	0.06	0.05	0.08	0.14	−0.20	−0.14	−0.05	0.86	0.55	0.92	−0.23	−0.33	−0.08
const.-unconst	(0.86)	(1.01)	(0.42)	(0.18)	(0.45)	(0.60)	(−0.74)	(−0.71)	(−0.24)	(3.31)	(3.82)	(4.13)	(−1.16)	(−2.27)	(−0.51)
High *RDS*,	0.57	0.44	0.42	0.56	0.54	0.50	0.51	0.45	0.41	1.36	1.18	1.25	0.57	0.19	0.49
const.-unconst.	(1.99)	(2.00)	(1.64)	(1.53)	(2.74)	(1.46)	(1.46)	(2.39)	(1.33)	(3.77)	(6.18)	(3.71)	(1.61)	(0.79)	(1.67)
Low *RDCAP*,	0.35	0.28	0.10	0.13	0.16	0.22	−0.26	−0.01	−0.16	0.70	0.39	0.74	−0.32	−0.39	−0.09
const.-unconst	(1.59)	(1.73)	(0.66)	(0.47)	(0.86)	(0.93)	(−0.86)	(−0.07)	(−0.62)	(2.67)	(2.60)	(3.21)	(−1.40)	(−2.48)	(−0.54)
High *RDCAP*,	0.71	0.45	0.57	0.51	0.51	0.45	0.58	0.44	0.49	1.43	1.21	1.34	0.61	0.13	0.52
const.-unconst.	(2.35)	(1.89)	(2.09)	(1.40)	(2.52)	(1.31)	(1.61)	(2.23)	(1.50)	(4.09)	(6.62)	(4.09)	(1.73)	(0.54)	(1.76)
Low *RDME*,	0.16	0.17	0.08	−0.25	−0.20	−0.19	−0.30	−0.25	−0.25	0.50	0.22	0.50	−0.07	−0.24	0.05
const.-unconst	(0.75)	(0.94)	(0.52)	(−0.87)	(−1.14)	(−0.79)	(−0.90)	(−1.19)	(−0.91)	(1.85)	(1.43)	(2.14)	(−0.22)	(−1.30)	(0.20)
High *RDME*,	0.45	0.20	0.30	0.82	0.87	0.80	0.86	0.83	0.82	1.49	1.29	1.43	0.70	0.28	0.70
const.-unconst.	(1.76)	(1.04)	(1.40)	(2.28)	(4.39)	(2.40)	(2.44)	(4.68)	(2.59)	(4.75)	(8.09)	(4.93)	(2.27)	(1.29)	(2.63)

(*continued*)

Table 4
Continued

	Panel A: KZ Index			Panel B: WW Index			Panel C: SA Index			Panel D: Size			Panel E: Age		
	Raw return	Charact.-adjusted return	Industry-adjusted return	Raw return	Charact.-adjusted return	Industry-adjusted return	Raw return	Charact.-adjusted return	Industry-adjusted return	Raw return	Charact.-adjusted return	Industry-adjusted return	Raw return	Charact.-adjusted return	Industry-adjusted return
Low *RDE*,	0.18	0.15	0.07	0.08	0.16	0.16	−0.23	0.00	−0.06	0.82	0.55	0.89	−0.23	−0.28	−0.05
const.-unconst.	(0.90)	(0.91)	(0.44)	(0.29)	(0.86)	(0.71)	(−0.86)	(−0.02)	(−0.27)	(3.27)	(3.85)	(3.93)	(−1.11)	(−1.81)	(−0.32)
High *RDE*,	0.49	0.39	0.30	0.51	0.52	0.50	0.47	0.41	0.42	1.40	1.25	1.32	0.51	0.12	0.45
const.-unconst.	(1.69)	(1.77)	(1.21)	(1.36)	(2.52)	(1.43)	(1.32)	(2.09)	(1.33)	(3.84)	(6.49)	(3.90)	(1.42)	(0.50)	(1.56)
Low *RDA*,	0.24	0.26	0.03	0.03	0.09	0.13	−0.30	−0.11	−0.13	0.74	0.44	0.81	−0.43	−0.48	−0.19
const.-unconst.	(1.21)	(1.60)	(0.23)	(0.12)	(0.50)	(0.55)	(−1.08)	(−0.58)	(−0.57)	(2.84)	(3.03)	(3.57)	(−1.78)	(−2.90)	(−1.05)
High *RDA*,	0.53	0.35	0.42	0.55	0.52	0.50	0.50	0.44	0.41	1.41	1.22	1.31	0.63	0.21	0.54
const.-unconst.	(1.71)	(1.57)	(1.54)	(1.49)	(2.64)	(1.43)	(1.42)	(2.30)	(1.30)	(4.00)	(6.71)	(3.96)	(1.72)	(0.84)	(1.79)

At the end of June of year t, I sort R&D-reporting firms with positive real sales growth into two R&D groups and three financial constraints groups independently. All sorting variables are for the fiscal year ending in year $t-1$ except size. The intersection of these groups forms six R&D-constraints portfolios, which are held for the next 12 months and reformed every year. I report the average value-weighted monthly return; characteristic-adjusted return by size, book-to-market, and momentum; and industry-adjusted return for the constrained-minus-unconstrained portfolios created in the low and high R&D groups. Heteroscedasticity-robust t-statistics are reported in parentheses. *RDCA* is R&D capital scaled by total assets. R&D capital is computed as the five-year cumulative R&D expenditures assuming an annual depreciation rate of 20%. *RDS*, *RDCAP*, *RDME*, *RDE*, and *RDA* are R&D expenditure scaled by sales, capital expenditure, year-end market equity, number of employees, and assets. The KZ index, the WW index, and the SA index are indices of financial constraints. Size is market capitalization at the end of June of year t. Age is the number of years a firm has been on Compustat with a non-missing stock price. The sample is from 1975 to 2007.

RDS, low RDCAP, low RDME, low RDE, and low RDA groups, respectively. This difference is similar for the SA index in Panel C of Table 4.

Panels D and E in Table 4, which contain results for size and firm age, confirm the model's predictions as well. The returns of the small-minus-big portfolios formed in the high R&D groups oftentimes more than double the returns of those formed in the low R&D groups. For example, the value-weighted monthly average return, the characteristic-adjusted return, and the industry-adjusted return of the small-minus-big portfolio are 1.45%, 1.22%, and 1.33%, with t-statistics of 4.13, 6.78, and 4.06, respectively, in the high RDCA group. In contrast, these estimates in the low RDCA group are only 0.72%, 0.42%, and 0.79%, with t-statistics of 2.70, 2.85, and 3.41, respectively. Similarly, the age-return relation in the high R&D groups is also much stronger than that in the low R&D groups. For example, the value-weighted return of the young-minus-old age portfolio formed in the high RDME group is 0.70%, with a t-statistic of 2.27. In contrast, the counterpart of this estimate is −0.07%, with a t-statistic of −0.22 in the low RDME group.

To examine whether extremely small firms drive the above findings, I sort firms independently into two R&D groups, two size groups, and three constraints groups. The intersection forms twelve constraints-size-R&D portfolios. Table 5 reports returns of the constrained-minus-unconstrained portfolios formed within the four R&D-size subsamples. The results show that the positive KZ-return relation and the age-return relation exist mainly among large R&D-intensive firms. For example, among big high RDCA firms, the value-weighted monthly return, the characteristic-adjusted return, and the industry-adjusted return of the high-minus-low KZ portfolio are 0.59%, 0.45%, and 0.47%, with t-statistics of 1.92, 1.95, and 1.74, respectively. In contrast, among small high RDCA firms, these estimates are only 0.10%, 0.13%, and 0.15%, with t-statistics of 0.48, 0.61, and 0.65, respectively.

Similarly, the returns of the high-minus-low WW portfolios and the high-minus-low SA portfolios formed among big high R&D firms are also higher than those formed among small high R&D firms, although they are statistically insignificant. The weakened results are likely due to the extremely high correlations between size and the WW and the SA indices. As discussed before, the Spearman's rank correlations between size and the WW index and the SA index are −0.82 and −0.74, respectively, whereas the correlations between size and the KZ index and age are −0.22 and 0.39, respectively. These patterns are robust to alternative measures of R&D and suggest that the positive constraints-return relation among high R&D firms is unlikely to be driven by extremely small firms, which are consistent with the FM regressions.[14]

To study what risk factor(s) may drive the positive constraints-return relation among high R&D firms, I regress the time-series returns of the

[14] To save space, I only report the results for *RDCA*, *RDS*, and *RDCAP*. The results are similar for the other three R&D measures.

Table 5
The returns of the constrained-minus-unconstrained portfolios across subsamples split by R&D and size

	Panel A: KZ Index			Panel B: WW Index			Panel C: SA Index			Panel D: Age		
	Raw return	Charact-adj return	Industry-adjusted return	Raw return	Charact.-adj return	Industry-adjusted return	Raw return	Charact.-adj return	Industry adjusted return	Raw return	Charact.-adj return	Industry-adjusted return
Low *RDCA*, small,	−0.10	−0.09	−0.16	−0.12	0.33	−0.02	−0.69	−0.18	−0.52	−0.75	−0.47	−0.61
const.-unconst	(−0.47)	(−0.42)	(−0.76)	(−0.35)	(1.03)	(−0.05)	(−2.67)	(−0.81)	(−2.17)	(−3.14)	(−2.04)	(−2.68)
Low *RDCA*, big,	0.17	0.17	0.00	−0.21	−0.29	−0.15	−0.58	−0.62	−0.50	−0.45	−0.48	−0.20
const.-unconst.	(0.81)	(0.99)	(0.01)	(−0.50)	(−0.82)	(−0.39)	(−1.38)	(−1.50)	(−1.32)	(−1.62)	(−2.46)	(−1.04)
High *RDCA*, small,	0.10	0.13	0.15	−0.24	−0.23	0.28	−1.11	−0.61	−1.13	−0.38	−0.07	−0.38
const-unconst	(0.48)	(0.61)	(0.65)	(−0.47)	(−0.44)	(−0.56)	(−3.37)	(−2.08)	(−3.50)	(−1.54)	(−0.31)	(−1.61)
High *RDCA*, big,	0.59	0.45	0.47	−0.04	−0.04	−0.06	0.32	−0.01	0.25	0.69	0.21	0.60
const-unconst	(1.92)	(1.95)	(1.74)	(−0.09)	(−0.13)	(−0.14)	(0.77)	(−0.05)	(0.66)	(1.72)	0.74	1.79
Low *RDS*, small,	−0.14	−0.12	−0.15	−0.02	0.42	0.01	−0.67	−0.21	−0.55	−0.79	−0.57	−0.67
const.-unconst	(−0.73)	(−0.65)	(−0.78)	(−0.07)	(1.35)	(0.04)	(−2.74)	(−0.98)	(−2.37)	(−3.32)	(−2.38)	(−2.95)
Low *RDS*, big,	0.20	0.18	0.07	−0.56	−0.25	−0.50	−1.07	−0.96	−0.94	−0.31	−0.39	−0.15
const.-unconst.	(0.91)	(1.03)	(0.45)	(−1.40	(−0.69)	(−1.36)	(−2.53)	(−2.21)	(−2.45)	(−1.50)	(−2.39)	(−0.94)
High *RDS*, small,	−0.01	0.01	0.02	−0.14	−0.60	−0.12	−0.92	−0.52	−0.91	−0.36	−0.04	−0.33
const.-unconst.	(−0.06)	(0.03)	(0.07)	(−0.24	(−0.94)	(−0.21)	(−2.79)	(−1.68)	(−2.87)	(−1.52)	(−0.19)	(−1.41)
High *RDS*, big,	0.60	0.44	0.44	0.09	−0.03	0.06	0.32	−0.01	0.25	0.54	0.12	0.46
const.-unconst.	(2.02)	(1.90)	(1.65)	(0.22)	(−0.12)	(0.15)	(0.82)	(−0.03)	(0.70)	(1.48)	(0.46)	(1.54)

(*continued*)

Table 5
Continued

	Panel A: KZ Index			Panel B: WW Index			Panel C: SA Index			Panel D: Age		
	Raw return	Charact-adj return	Industry-adjusted return	Raw return	Charact.-adj return	Industry-adjusted return	Raw return	Charact.-adj return	Industry adjusted return	Raw return	Charact.-adj return	Industry-adjusted return
Low *RDCAP*,small,	−0.11	−0.10	−0.14	−0.13	0.25	−0.09	−0.55	−0.13	−0.38	−0.68	−0.49	−0.49
const.-unconst.	(−0.52)	(−0.47)	(−0.70)	(−0.38)	(0.82)	(−0.28)	(−2.04)	(−0.58)	(−1.56)	(−2.78)	(−2.09)	(−2.08)
Low *RDCAP*, big,	0.36	0.29	0.11	−0.42	−0.31	−0.33	−0.94	−0.51	−0.79	−0.42	−0.47	−0.18
const.-unconst.	(1.62)	(1.75)	(0.69)	(−1.11)	(−0.94)	(−1.01)	(−2.17)	(−1.27)	(−1.99)	(−1.79)	(−2.71)	(−0.98)
High *RDCAP*, small,	0.10	0.14	0.16	−0.55	−0.50	−0.47	−1.02	−0.47	−1.02	−0.47	−0.05	−0.50
const.-unconst.	(0.45)	(0.64)	(0.65)	(−1.06)	(−0.98)	(−0.90)	(−3.19)	(−1.89)	(−3.19)	(−1.89)	(−0.20)	(−2.08)
High *RDCAP*, big,	0.69	0.39	0.54	0.05	0.00	0.03	0.41	−0.12	0.34	0.59	0.05	0.50
const.-unconst.	(2.20)	(1.53)	(1.92)	(0.10)	(0.00)	(0.07)	(1.01)	(−0.43)	(0.92)	(1.63)	(0.19)	(1.67)

At the end of June of year t, I sort R&D-reporting firms with positive real sales growth into two R&D groups, two size groups, and three financial constraints groups independently. All sorting variables are for the fiscal year ending in year $t-1$ except size. The intersection of these groups forms twelve R&D-size-constraints portfolios, which are held for the next 12 months and reformed every year. I report the average value-weighted monthly return; characteristic-adjusted return by size, book-to-market, and momentum; and industry-adjusted return for the constrained-minus-unconstrained portfolios created in the four R&D-size groups. Heteroscedasticity-robust t-statistics are reported in parentheses. *RDCA* is R&D capital scaled by total assets. R&D capital is the five-year cumulative R&D expenditures, assuming an annual depreciation rate of 20%. *RDS* and *RDCAP* are R&D expenditure scaled by sales and capital expenditure. The KZ index, the WW index, and the SA index are indices of financial constraints. Age is the number of years a firm has been on Compustat with a non-missing stock price. The sample is for 1975–2007.

constrained-minus-unconstrained portfolios on the market, size, book-to-market, momentum, and liquidity factors. The market, size, and book-to-market factors are detailed in Fama and French (1993). The momentum and liquidity factors are detailed in Jegadeesh and Titman (1993, 2001) and Pastor and Stambaugh (2003), respectively.

Table 6 shows that the relation is mainly driven by the market and size factors. For example, in the high RDCA subsample, the loadings of the high-minus-low KZ portfolios on the market and size factors are positive and significant, 0.33 and 0.76, with t-statistics of 5.10 and 7.82, respectively. In general, the loadings of the hedge portfolios on the book-to-market factor are negative, and the loadings on the momentum and liquidity factors are insignificant. Furthermore, the returns of these hedge portfolios generally can be fully explained by these factors. The pattern is similar for size and age in unreported results.

Financial constraints are related to financial distress to a certain degree. One concern is whether the leverage effect or associated bankruptcy risk drives the relation between financial constraints and stock returns observed in high R&D firms. To answer this question, I sort firms on R&D intensity first and then on leverage ratio, defined as the ratio of total debt (Compustat item 9 plus item 34) to total capital (total debt plus Compustat item 216). I find the relation between the leverage ratio and expected stock returns flat. In fact, high R&D firms have the lowest leverage ratio, which can be due to these firms' low debt capacity and unwillingness to borrow. Therefore, the leverage effect is unlikely to cause the strong positive relation between financial constraints and expected stock returns among R&D-intensive firms. In addition, the sample excludes R&D firms with negative real sales growth, which also helps reduce the confounding effect of financial distress.

Another concern is whether these findings are unique to R&D-intensive firms. As explained earlier, financial constraints affect R&D investment decisions more than capital investment decisions since R&D is much less flexible to adjustments. Insufficient funds usually lead to suspension of R&D projects. Furthermore, suspension affects R&D-intensive firms more than capital-intensive firms due to the high uncertainty and the intense competition in R&D-intensive industries. Therefore, the positive constraints-return relation should manifest itself most strongly in R&D-intensive firms.

To verify this intuition, I examine the KZ-return relation across different levels of capital investment measured by the ratio of capital expenditure to PPE (plant, property, and equipment). I find the relation flat. This result contrasts sharply with the strong positive KZ-return relation among R&D-intensive firms. Hence, R&D-intensive firms provide a good framework for identifying the asset-pricing implication of financial constraints.

2.3.2 Variation of the R&D-return relation with financial constraints. Table 7 shows that the positive R&D-return relation exists only among financially constrained firms. For example, among small firms, the value-weighted

Table 6
The risk factor loadings of the constrained-minus-unconstrained portfolios across subsamples split by R&D

	KZ Index					WW Index					SA Index				
	Mkt	*SMB*	*HML*	textitMom	*Liq*	*Mkt*	*SMB*	*HML*	*Mom*	*Liq*	*Mkt*	*SMB*	*HML*	*Mom*	*Liq.*
Low *RDCA*,	0.07	0.08	0.26	0.22	−0.07	0.13	1.23	0.00	−0.05	−0.04	0.17	0.98	−0.37	−0.10	−0.09
const-unconst	(1.17)	(0.84)	(2.24)	(3.02)	(−1.38)	(2.36)	(13.91)	(−0.05)	(−0.89)	(−0.77)	(2.31)	(7.83)	(−3.15)	(−1.18)	(−1.11)
High *RDCA*,	0.33	0.76	−0.30	0.17	−0.08	0.23	1.62	−0.34	0.01	0.06	0.29	1.60	−0.39	0.00	0.09
const-unconst	(5.10)	(7.82)	(−2.42)	(2.16)	(−1.37)	(4.03)	(17.13)	(−3.54)	(0.18)	(1.02)	(5.00)	(18.10)	(−4.23)	(−0.06)	(1.57)
Low *RDS*,	0.14	0.15	−0.06	0.12	0.02	0.14	1.19	0.13	−0.13	−0.07	0.16	1.07	0.05	−0.08	−0.09
const-unconst	(2.21)	(1.46)	(−0.50)	(1.58)	(0.29)	(2.39)	(13.72)	(1.45)	(−2.14)	(−1.06)	(2.56)	(10.15)	(0.55)	(−1.28)	(−1.22)
High *RDS*,	0.29	0.78	−0.19	0.11	−0.06	0.22	1.59	−0.33	0.05	0.05	0.29	1.45	−0.38	0.00	0.03
const-unconst	(4.26)	(8.49)	(−1.53)	(1.46)	(−0.94)	(3.81)	(15.80)	(−3.48)	(0.66)	(0.87)	(4.93)	(17.17)	(−4.12)	(0.02)	(0.66)
Low *RDCAP*,	0.15	0.25	−0.09	0.11	0.01	0.10	1.20	−0.11	−0.03	−0.10	0.12	1.08	−0.27	−0.14	−0.07
const-unconst	(2.50)	(2.35)	(−0.81)	(1.25)	(0.27)	(1.80)	(12.43)	(−1.25)	(−0.47)	(−1.68)	(1.72)	(8.54)	(−2.34)	(−1.60)	(−0.91)
High *RDCAP*,	0.29	0.89	−0.13	0.23	−0.16	0.26	1.63	−0.18	0.00	0.08	0.31	1.58	−0.29	0.04	0.09
const-unconst	(4.16)	(9.68)	(−1.01)	(2.80)	(−2.73)	(4.10)	(14.88)	(−1.76)	(0.00)	(1.24)	(4.82)	(17.93)	(−2.89)	(0.51)	(1.46)
Low *RDME*,	0.18	0.23	0.24	0.12	−0.02	0.18	1.27	−0.08	−0.13	−0.11	0.26	1.34	−0.39	−0.11	0.02
const-unconst	(3.12)	(2.18)	(2.25)	(1.46)	(−0.45)	(3.37)	(14.73)	(−0.95)	(−2.09)	(−1.90)	(4.35)	(13.50)	(−3.63)	(−1.68)	(0.39)
High *RDME*,	0.29	0.49	0.07	0.17	−0.14	0.14	1.41	−0.57	0.02	0.04	0.15	1.42	−0.62	0.03	0.05
const-unconst	(3.94)	(4.28)	(0.55)	(1.89)	(−1.65)	(2.27)	(14.56)	(−5.19)	(0.26)	(0.58)	(2.51)	(19.53)	(−6.14)	(0.55)	(0.92)
Low *RDE*,	0.11	0.18	−0.05	0.11	0.00	0.17	1.11	0.03	−0.07	−0.05	0.19	0.98	−0.12	−0.05	−0.04
const-unconst	(1.76)	(1.93)	(−0.47)	(1.57)	(0.04)	(3.00)	(13.20)	(0.30)	(−1.12)	(−0.80)	(3.09)	(9.00)	(−1.23)	(−0.83)	(−0.55)
High *RDE*,	0.33	0.65	−0.08	0.14	0.03	0.19	1.66	−0.37	−0.03	0.07	0.26	1.52	−0.43	−0.07	0.03
const-unconst	(4.95)	(6.24)	(−0.61)	(1.54)	(0.40)	(3.31)	(15.89)	(−3.94)	(−0.48)	(0.99)	(4.44)	(17.60)	(−4.75)	(−1.08)	(0.52)
Low *RDA*,	0.07	0.13	0.12	0.23	−0.02	0.14	1.24	0.11	−0.10	−0.04	0.14	1.13	−0.15	−0.09	−0.03
const.-unconst.	(1.24)	(1.46)	(1.12)	(3.60)	(−0.45)	(2.40)	(18.45)	(1.27)	(−1.89)	(−0.71)	(2.33)	(12.67)	(−1.55)	(−1.43)	(−0.47)
High *RDA*,	0.34	0.77	−0.31	0.18	−0.09	0.23	1.62	−0.33	0.02	0.06	0.30	1.48	−0.36	−0.02	0.04
const.-unconst.	(5.04)	(7.63)	(−2.46)	(2.16)	(−1.45)	(3.91)	(16.66)	(−3.42)	(0.29)	(0.96)	(4.99)	(17.51)	(−3.90)	(−0.32)	(0.67)

At the end of June of year t, I sort R&D-reporting firms with positive real sales growth into two R&D groups and three financial constraints groups independently. The intersection of these groups forms six R&D-constraints portfolios, which are held for the next 12 months and reformed every year. I also form a constrained-minus-unconstrained portfolio in each R&D groups. *RDCA* is R&D capital scaled by total assets. R&D capital is the five-year cumulative R&D expenditures, assuming an annual depreciation rate of 20%. *RDS*, *RDCAP*, *RDME*, *RDE*, and *RDA* are R&D expenditure scaled by sales, capital expenditure, year-end market equity, number of employees, and assets. The KZ index, the WW index, and the SA index are indices of financial constraints. All sorting variables are for the fiscal year ending in year $t-1$. I regress value-weighted monthly excess returns of the constrained-minus-unconstrained portfolios on the market, size, book-to-market, momentum, and liquidity factors. I report the slopes on these factors and the heteroscedasticity-robust t-statistics in parentheses. *Mkt*, *SMB*, *HML*, *Mom*., and *Liq*. are the market, size, book-to-market, momentum, and liquidity factors. The sample is from 1975 to 2007.

Table 7
The returns of the high-minus-low R&D portfolios across subsamples split by financial constraints

	RDCA			*RDCAP*			*RDA*			*RDME*			*RDS*			*RDE*		
	Raw return	Char-adj. retn	Ind-adj. rtn	Raw return	Char-adj. retn	Ind-adj. rtn	Raw return	Char-adj. retn	Ind-adj. rtn	Raw return	Char-adj. retn	Ind-adj. retn	Raw return	Char-adj. retn	Ind-adj. retn	Raw return	Char-adj. retn	Ind-adj. retn
Big size,	0.03	0.12	0.16	−0.04	0.04	0.10	−0.03	0.09	0.11	0.19	0.11	0.23	−0.08	0.02	0.08	−0.05	0.07	0.07
H-L R&D	(0.15)	(1.01)	(2.12)	(−0.23)	(0.39)	(1.06)	(−0.18)	(0.80)	(1.38)	(1.11)	(0.88)	(2.22)	(−0.42)	(0.20)	(0.98)	(−0.27)	(0.64)	(0.87)
Small size,	0.76	0.92	0.71	0.69	0.86	0.69	0.64	0.87	0.61	1.18	1.19	1.16	0.43	0.65	0.42	0.53	0.78	0.51
H-L R&D	(2.93)	(3.69)	(3.16)	(2.64)	(3.45)	(3.12)	(2.38)	(3.49)	(2.67)	(4.88)	(5.21)	(5.65)	(1.50)	(2.50)	(1.74)	(1.96)	(3.02)	(2.17)
Old age,	−0.14	0.05	0.04	−0.14	0.00	0.04	−0.17	0.03	0.01	0.11	0.09	0.14	−0.17	−0.02	0.01	−0.12	0.05	0.03
H-L R&D	(−0.79)	(0.40)	(0.51)	(−0.85)	(0.04)	(0.53)	(−0.94)	(0.24)	(0.11)	(0.65)	(0.71)	(1.35)	(−0.95)	(−0.13)	(0.10)	(−0.73)	(0.45)	(0.31)
Young age,	0.94	0.71	0.79	0.80	0.52	0.65	0.89	0.71	0.73	0.89	0.61	0.79	0.64	0.51	0.57	0.62	0.45	0.52
H-L R&D	(2.91)	(2.29)	(2.87)	(2.52)	(1.94)	(2.60)	(2.85)	(2.53)	(2.92)	(3.30)	(2.39)	(3.41)	(1.76)	(1.69)	(2.13)	(1.79)	(1.52)	(1.98)
Low SA,	−0.04	0.09	0.12	−0.08	0.03	0.07	−0.08	0.07	0.08	0.14	0.09	0.19	−0.11	0.01	0.06	−0.08	0.05	0.04
H-L R&D	(−0.23)	(0.78)	(1.52)	(−0.49)	(0.24)	(0.79)	(−0.45)	(0.60)	(0.97)	(0.84)	(0.76)	(1.85)	(−0.64)	(0.04)	(0.68)	(−0.51)	(0.47)	(0.52)
High SA,	0.69	0.61	0.63	0.76	0.48	0.73	0.72	0.62	0.62	1.30	1.17	1.26	0.59	0.60	0.52	0.62	0.47	0.52
H-L R&D	(2.19)	(2.01)	(2.09)	(2.37)	(1.81)	(2.39)	(2.52)	(2.12)	(2.60)	(5.43)	(4.94)	(5.44)	(1.87)	(1.99)	(2.04)	(2.06)	(1.60)	(2.08)
Low KZ,	0.08	0.21	0.09	0.18	0.26	0.08	0.13	0.27	0.08	0.25	0.18	0.30	0.01	0.13	0.08	0.04	0.21	0.09
H-L R&D	(0.38)	(1.20)	(0.58)	(0.93)	(1.61)	(0.54)	(0.64)	(1.68)	(0.56)	(1.20)	(1.10)	(1.70)	(0.06)	(0.77)	(0.49)	(0.22)	(1.29)	(0.58)
High KZ,	0.52	0.51	0.58	0.55	0.43	0.55	0.41	0.36	0.47	0.53	0.21	0.53	0.40	0.40	0.43	0.35	0.45	0.33
H-L R&D	(1.62)	(2.02)	(2.44)	(1.73)	(1.62)	(2.22)	(1.26)	(1.42)	(1.92)	(1.96)	(0.89)	(2.61)	(1.27)	(1.51)	(1.86)	(1.19)	(1.83)	(1.44)
Low WW,	−0.01	0.09	0.14	−0.06	0.04	0.09	−0.05	0.08	0.10	0.15	0.08	0.20	−0.10	0.01	0.07	−0.06	0.06	0.06
H-L R&D	(−0.04)	(0.79)	(1.80)	(−0.33)	(0.32)	(1.02)	(−0.29)	(0.71)	(1.20)	(0.83)	(0.64)	(1.91)	(−0.54)	(0.06)	(0.84)	(−0.35)	(0.51)	(0.69)
High WW,	0.42	0.52	0.40	0.32	0.38	0.33	0.46	0.51	0.47	1.22	1.15	1.19	0.41	0.47	0.44	0.37	0.41	0.40
H-L R&D	(1.37)	(1.73)	(1.44)	(1.04)	(1.29)	(1.16)	(1.42)	(1.66)	(1.63)	(4.36)	(4.23)	(4.56)	(1.19)	(1.48)	(1.44)	(1.11)	(1.35)	(1.34)

At the end of June of year t, I sort R&D-reporting firms with positive real sales growth into two R&D groups and three financial constraints groups independently. All sorting variables are for the fiscal year ending in year $t-1$ except size. The intersection of these groups forms six R&D-constraints portfolios, which are held for the next 12 months and reformed every year. I report the average value-weighted monthly return; characteristic-adjusted return (Char.-adj. retn) by size, book-to-market, and momentum; and industry adjusted return (Ind.-adj. retn) for the high-minus-low R&D portfolios created in the financial constraints groups. Heteroscedasticity-robust t-statistics are reported in parentheses. *RDCA* is R&D capital scaled by total assets. I compute R&D capital as the five-year cumulative R&D expenditures assuming an annual depreciation rate of 20% *RDCAP*, *RDA*, *RDME*, *RDS*, and *RDE* are R&D expenditure scaled by capital expenditure, total assets, year-end market equity, sales, and number of employees. Size is market capitalization at the end of June of year t. Age is the number of years a firm has been on Compustat with a non-missing stock price. The SA index, the KZ index, and the WW index are indices of financial constraints. The sample is for 1975–2007.

monthly return, the characteristic-adjusted return, and the industry-adjusted return of the high-minus-low RDCAP portfolio are 0.69%, 0.86%, and 0.69%, respectively, and all are significant at the 1% level. In contrast, among big firms, these estimates are only −0.04%, 0.04%, and 0.10%, respectively, and none of them are significant.

This sharp contrasting result that the positive R&D-return relation exists only among financially constrained firms is robust to alternative measures of R&D intensity and financial constraints such as age and the SA index. For example, the corresponding return estimates of the high-minus-low RDS portfolio in the high SA subsample are 0.59%, 0.60%, and 0.52%, with t-statistics of 1.87, 1.99, and 2.04, respectively. However, these estimates in the low SA subsample are only −0.11%, 0.01%, and 0.06%, with t-statistics of −0.64, 0.04, and 0.68, respectively. Similarly, the returns of the high-minus-low R&D portfolios in the high KZ and high WW subsamples are also much higher than those in the low KZ and low WW subsamples, although sometimes they are statistically insignificant. Furthermore, although several measures of R&D intensity cannot predict returns in the single sort using the whole sample and value-weighted returns, they can do so among constrained firms such as small, young, and high SA firms. These findings suggest that financial constraints potentially drive the positive R&D-return relation.

Finally, I also investigate what systematic risk(s) may drive the R&D-return relation by regressing the high-minus-low R&D portfolio returns on standard risk factors. Table 8 shows constrained high R&D firms load significantly higher on the size factor than constrained low R&D firms. Sometimes constrained high R&D firms also load significantly higher on the market and liquidity factors. These findings suggest that increased risks induced by financial constraints contribute to the positive R&D-return relation.

3. Conclusion

Two puzzles in the literature have attracted a fair amount of attention: the mixed evidence on the relation between financial constraints and stock returns, and the positive R&D-return relation. This article provides new perspectives by studying these relations via the interaction between financial constraints and R&D investment. Unlike capital investment, R&D investment is less flexible. A financially constrained R&D-intensive firm is more likely to suspend/ discontinue its R&D projects. Therefore, R&D-intensive firms' risk increases with their financial constraints status. Conversely, financially constrained firms' risk increases with their R&D intensity. I find that a robust constraints-return relation exists among R&D-intensive firms, and the positive R&D-return relation exists only among financially constrained firms. These findings suggest that financial constraints have a significant impact on R&D-intensive firms' risk and return and potentially drive the positive R&D-return relation.

Table 8
The risk factor loadings of the high-minus-low R&D portfolios across subsamples split by financial constraints

	RDCTA					RDCAP					RDS					RDME				
	Mkt	*SMB*	*HML*	*Mom*	*Liq.*	*Mkt*	*SMB*	*HML*	*Mom*	*Liq.*	*Mkt*	*SMB*	*HML*	*Mom*	*Liq.*	*Mkt*	*SMB*	*HML*	*Mom*	*Liq.*
Big size,	−0.08	0.11	−0.43	−0.09	0.06	−0.05	0.10	−0.62	−0.11	0.08	−0.04	0.17	−0.54	−0.16	0.11	0.02	0.48	0.22	−0.04	0.06
H-L R&D	(−1.74)	(1.43)	(−5.43)	(−1.28)	(1.51)	(−1.26)	(1.52)	(−8.18)	(−1.67)	(2.62)	(−0.81)	(2.18)	(−6.66)	(−2.23)	(3.06)	(0.29)	(5.11)	(2.17)	(−0.46)	(1.11)
Small size,	0.09	0.60	−0.53	−0.07	0.12	0.11	0.59	−0.53	−0.08	0.18	0.14	0.69	−0.63	−0.08	0.11	0.11	0.56	−0.32	−0.02	0.15
H-L R&D	(1.65)	(5.83)	(−6.20)	(−1.17)	(1.98)	(2.09)	(4.94)	(−5.94)	(−1.30)	(2.43)	(2.15)	(6.45)	(−6.65)	(−1.15)	(1.77)	(1.88)	(4.91)	(−3.68)	(−0.29)	(2.22)
Old age,	−0.08	0.10	−0.41	−0.11	0.06	−0.09	0.05	−0.53	−0.15	0.05	−0.08	0.11	−0.41	−0.13	0.07	0.05	0.48	0.18	−0.08	0.09
H-L R&D	(−1.73)	(1.25)	(−4.81)	(−1.67)	(1.70)	(−1.95)	(0.71)	(−7.06)	(−2.70)	(1.58)	(−1.60)	(1.41)	(−4.91)	(−1.99)	(1.88)	(0.96)	(4.99)	(1.96)	(−0.96)	(1.60)
Young age,	0.13	0.52	−0.64	−0.12	0.06	0.16	0.31	−0.97	−0.03	0.06	0.21	0.31	−1.30	−0.15	0.10	−0.07	0.56	0.10	0.12	−0.08
H-L R&D	(1.60)	(2.79)	(−4.87)	(−1.32)	(0.47)	(2.01)	(2.59)	(−7.09)	(−0.28)	(0.87)	(2.59)	(2.86)	(−9.31)	(−1.39)	(1.48)	(−0.89)	(5.05)	(0.74)	(1.48)	(−1.06)
Low SA,	−0.08	0.07	−0.43	−0.10	0.06	−0.07	0.04	−0.57	−0.13	0.05	−0.06	0.10	−0.47	−0.16	0.08	0.02	0.46	0.21	−0.04	0.05
H-L R&D	(−1.70)	(0.99)	(−5.24)	(−1.37)	(1.64)	(−1.67)	(0.64)	(−7.56)	(−1.97)	(1.83)	(−1.31)	(1.39)	(−5.92)	(−2.34)	(2.33)	(0.46)	(4.97)	(2.09)	(−0.47)	(0.88)
High SA,	0.05	0.69	−0.45	0.00	0.24	0.12	0.54	−0.59	0.04	0.21	0.07	0.48	−0.90	−0.08	0.21	−0.09	0.54	−0.02	0.10	0.08
H-L R&D	(0.69)	(4.40)	(−3.34)	(0.01)	(2.38)	(1.58)	(3.78)	(−3.97)	(0.41)	(2.56)	(1.10)	(3.70)	(−8.42)	(−0.94)	(2.64)	(−1.57)	(4.67)	(−0.19)	(1.41)	(1.22)
Low KZ,	−0.16	−0.22	−0.21	0.12	−0.03	−0.08	−0.03	−0.59	−0.02	0.08	−0.10	−0.09	−0.52	−0.01	0.09	−0.06	0.41	0.07	−0.06	0.12
H-L R&D	(−2.83)	(−2.12)	(−1.96)	(1.72)	(−0.53)	(−1.62)	(−0.48)	(−7.06)	(−0.42)	(2.06)	(−1.72)	(−0.97)	(−5.36)	(−0.23)	(1.75)	(−0.96)	(3.80)	(0.65)	(−0.79)	(1.49)
High KZ,	0.09	0.46	−0.77	0.08	−0.04	0.05	0.61	−0.63	0.10	−0.09	0.05	0.54	−0.66	−0.02	0.02	0.05	0.67	−0.10	−0.01	0.01
H-L R&D	(1.14)	(3.61)	(−6.13)	(0.74)	(−0.54)	(0.62)	(5.25)	(−4.96)	(0.96)	(−1.34)	(0.65)	(4.44)	(−5.14)	(−0.24)	(0.20)	(0.64)	(6.92)	(−0.88)	(−0.11)	(0.15)
Low WW,	−0.10	0.11	−0.38	−0.09	0.04	−0.08	0.07	−0.58	−0.11	0.06	−0.06	0.14	−0.48	−0.16	0.09	0.03	0.44	0.24	−0.06	0.05
H-L R&D	(−2.13)	(1.43)	(−4.73)	(−1.26)	(1.13)	(−1.77)	(1.10)	(−7.60)	(−1.64)	(1.96)	(−1.30)	(1.96)	(−5.90)	(−2.26)	(2.63)	(0.63)	(4.73)	(2.32)	(−0.65)	(0.87)
High WW,	0.00	0.50	−0.72	−0.02	0.15	0.08	0.50	−0.64	−0.09	0.24	0.02	0.54	−0.94	0.01	0.22	−0.01	0.59	−0.25	0.09	0.19
H-L R&D	(0.04)	(3.57)	(−6.35)	(−0.28)	(1.76)	(1.24)	(3.32)	(−5.81)	(−1.11)	(2.80)	(0.25)	(3.36)	(−7.82)	(0.15)	(2.13)	(−0.13)	(3.52)	(−2.11)	(0.96)	(1.80)

At the end of June of year t, I sort R&D-reporting firms with positive real sales growth into two R&D groups and three financial constraints groups independently. The intersection of these groups forms six R&D-constraints portfolios, which are held for the next 12 months and reformed every year. I also form a high-minus-low R&D portfolio in each financial constraints group. *RDCA* is R&D capital scaled by total assets. I compute R&D capital as the five-year cumulative R&D expenditures assuming an annual depreciation rate of 20%. *RDCAP*, *RDS*, and *RDME* are R&D expenditure scaled by capital expenditure, sales, and year-end market equity. Size is market capitalization at the end of June of year t. Age is the number of years a firm has been on Compustat with a non-missing stock price. The SA index, the KZ index, and the WW index are indices of financial constraints. All sorting variables are for the fiscal year ending in year $t-1$ except size. I regress the value-weighted monthly excess returns of the high-minus-low R&D portfolios on the market, size, book-to-market, momentum, and liquidity factors. I report the slopes on these factors and the heteroscedasticity-robust t-statistics in parentheses. *Mkt*, *SMB*, *HML*, *Mom*., and *Liq*. are the market, size, book-to-market, momentum, and liquidity factors. The sample is for 1975–2007.

Appendix A: Proofs

Solutions and Proof of Valuation Functions. For $n < N$, let $V^c(y,n)$ solve Equation (3) with $\upsilon = 1$ and let $V^m(y,n)$ solve Equation (3) with $\upsilon = 0$. Let these functions satisfy the single-crossing property in y. Then, for every n, a $y^*(n)$ exists such that

$$V(y,n) = \begin{cases} V^m(y,n) = D(n)y^{\beta} & y < y^*(n) \\ V^c(y,n) = \sum_{i=n}^{N-1} C(i,n)y^{\gamma(i)} + B(n)y + A(n) & y \geq y^*(n) \end{cases} \tag{A1}$$

with

$$V(0,n) = 0 \tag{A2}$$

$$\lim_{y\to\infty} \frac{V(y,n)}{y} < \infty, \tag{A3}$$

where $\beta > 1$ and $\gamma(i) < 0$. When $y = y^*(n)$, the following boundary conditions hold:

$$V^c(y^*(n),n) = V^m(y^*(n),n) \tag{A4}$$

$$\frac{\partial}{\partial y} V^c(y^*(n),n) = \frac{\partial}{\partial y} V^m(y^*(n),n) \tag{A5}$$

$$\pi(n)[V(y^*_{CB}(n), n+1) - V(y^*_{CB}(n),n)] = x(n) \tag{A6}$$

$$p(n)\frac{1}{dt}E_t[dV(y^*_{FC}(n),n)] = x(n), \tag{A7}$$

where $y^*(n) = \max(y^*_{CB}(n), y^*_{FC}(n))$. I detail the constants β, $D(n)$, $\gamma(i)$, $C(i,n)$, $B(n)$, and $A(n)$ and the derivation of $y^*_{CB}(n)$ and $y^*_{FC}(n)$ below.

According to Dixit and Pindyck (1994), two conditions need to be satisfied for the single-crossing property. First, the difference between the benefit from continuing and the profit flow from suspension, which is zero in this case, increases with future cash flow. This condition is satisfied since the benefit from continuing is the expected value of future cash flow minus the expected investment cost. Second, this advantage from continuing will not reverse in the near term. Since future cash flow is a geometric Brownian motion and exhibits positive serial correlation, this condition is also satisfied in this model.

The condition $V(0,n) = 0$ is derived from the assumption that future cash flow $y(t)$ follows a geometric Brownian motion. Since $y(t)$ stays at zero forever if it ever reaches zero, the firm value at $y(t) = 0$ has to be 0. The condition $\lim_{y\to\infty} \frac{V(y,n)}{y} < \infty$ ensures that the firm value increases in proportion to future cash flow to prevent bubbles.

In the mothball region ($\upsilon = 0$), the HJB equation reduces to a homogeneous ordinary differential equation (ODE) with a standard solution given by

$$V^m(y,n) = D(n)y^{\beta} \qquad y < y^*(n), \tag{A8}$$

where β satisfies

$$\beta = \frac{(\sigma^2 - 2\mu) + \sqrt{8\sigma^2 r + (2\mu - \sigma^2)^2}}{2\sigma^2} > 1 \tag{A9}$$

and the boundary conditions determine the constant $D(n)$, which is detailed in Proposition 1.

By substituting the value function in the continuation region of Equation (A1) into the HJB equation and collecting terms, we easily show that this solution satisfies the HJB equation and the constants are given by

$$C(i,n) = \begin{cases} 0 & i = N \\ \frac{\pi(n)C(i,n+1)}{\pi(n)-\pi(i)} & n < i < N \end{cases}$$

$$\gamma(n) = \frac{(\sigma^2 - 2\mu) - \sqrt{8\sigma^2(r+\pi(n)) + (2\mu - \sigma^2)^2}}{2\sigma^2} < 0$$

$$B(n) = \frac{\pi(n)B(n+1)}{r+\pi(n)-\mu} > 0 \text{ with } B(N) = \frac{1}{r-\mu}$$

$$A(n) = \frac{\pi(n)A(n+1) - x(n)}{r+\pi(n)} < 0 \text{ with } A(N) = 0.$$

The boundary conditions determine the constant, $C(n,n)$, which is detailed in Proposition 1.

To determine the threshold $y^*(n)$, the constants $D(n)$ and $C(n,n)$, and the firm value, I need the boundary conditions. Equation (A4) is the familiar value matching (VM) condition by continuity of the value function, and Equation (A5) is the smooth pasting (SP) condition to ensure that the slopes of the two value functions match at the threshold $y^*(n)$. Equation (A6) is the cost-benefit (CB) condition, where the threshold $y^*_{CB}(n)$ is derived from the first three boundary conditions. The financial constraints (FC) condition is stated in Equation (A7), where the threshold $y^*_{FC}(n)$ is derived from Equations (A4), (A5), and (A7). Since the firm needs to satisfy both financial constraints and cost benefit analysis, the threshold $y^*(n) = \max(y^*_{CB}(n), y^*_{FC}(n))$.

If $y^*_{FC}(n) > y^*_{CB}(n)$, the threshold $y^*(n)$ equals $y^*_{FC}(n)$. The threshold and the constants, $C(n,n)$ and $D(n)$, are derived from the boundary conditions stated in Equations (A4), (A5), and (A7). Utilizing Equation (1) at the optimum ($\upsilon = 1$), I can simplify Equation (A7) as follows:

$$p(n)\frac{1}{dt}E_t[dV] = p(n)[rV(y,n) + x(n)] = x(n),$$

which is equivalent to

$$k(n)V(n) = x(n),$$

where $k(n) \equiv \frac{p(n)r}{1-p(n)}$.

Since three unknowns and three equations exist, we easily verify that the threshold $y^*(n)$ solves the following equation:

$$\left[\sum_{i=n+1}^{N-1} C(i,n)(\gamma(i)-\gamma(n))(y^*(n))^{\gamma(i)}\right] + B(n)y^*(n)(1-\gamma(n)) - A(n)\gamma(n)$$

$$= (\beta - \gamma(n))\frac{x(n)}{k(n)},$$

and the constants $D(n)$ and $C(n,n)$ are given by

$$C(n,n) = (\gamma(n)-\beta)^{-1}(y^*(n))^{-\gamma(n)}[\sum_{i=n+1}^{N-1} C(i,n)(\beta-\gamma(i))(y^*(n))^{\gamma(i)}$$

$$+ B(n)y^*(n)(\beta-1) + \beta A(n)]$$

$$D(n) = \frac{x(n)}{k(n)}(y^*(n))^{-\beta}.$$

When $y^*_{FC}(n) < y^*_{CB}(n)$, the threshold $y^*(n)$ equals $y^*_{CB}(n)$. The threshold and the constants, $C(n,n)$ and $D(n)$, are derived from the boundary conditions stated in Equations (A4), (A5), and

(A6). It is easy to verify that the threshold $y^*(n)$ satisfies the following equation:

$$\sum_{i=n+1}^{N-1}[\gamma(i)\pi(n)-\gamma(n)\pi(i)+\beta(\pi(i)-\pi(n))]C(i,n)(y^*(n))^{\gamma(i)}=$$

$$B(n)y^*(n)\left[(r-\mu)(\beta-\gamma(n))+(\beta-1)\pi(n)\right]+A(n)\left(\beta\pi_n+r(\beta-\gamma(n))\right),$$

and the constants $D(n)$ and $C(n,n)$ are given by

$$C(n,n)=\frac{(y^*(n))^{-\gamma(n)}}{\pi(n)}\left[(r-\mu)B(n)y^*(n)+rA(n)-\sum_{i=n+1}^{N-1}\pi(i)C(i,n)(y^*(n))^{\gamma(i)}\right]$$

$$D(n)=\frac{(y^*(n))^{-\beta}}{\beta\pi(n)}\{(\pi(n)+(r-\mu)\gamma(n))B(n)y^*(n)+r\gamma(n)A(n)$$
$$+\sum_{i=n+1}^{N-1}(\gamma(i)\pi(n)-\gamma(n)\pi(i))C(i,n)(y^*(n))^{\gamma(i)}\}.$$

■

Proof of Proposition 1. When $n=N-1$, from Proposition 1 it can be easily shown that $y^*_{FC}(n)>y^*_{CB}(n)$ implies $(\beta-\gamma(n))(\beta-1)(r+\pi(n))+k(n)\gamma(n)>0$. For simplicity, the number of completed stages, n, is subsumed henceforth.

It is also easy to show $sign(\frac{\partial R}{\partial p})=sign(\frac{\partial R}{\partial k})$ and $sign(\frac{\partial^2 R}{\partial p\partial x})=sign(\frac{\partial^2 R}{\partial k\partial x})$. Taking the derivative of R with respect to k gives

$$\frac{\partial R}{\partial k}=\frac{\frac{\partial C(n,n)}{\partial k}y^\gamma[\gamma V-V_y y]}{V^2}\lambda.$$

After simplifying, I obtain

$$\frac{\partial C(n,n)}{\partial k}=\frac{x(y^*)^{-\gamma}}{k^2[(\beta-\gamma)(r+\pi)-k\gamma]}[(\beta-\gamma)(\beta-1)(r+\pi)+k\gamma]>0.$$

Therefore, $\frac{\partial R}{\partial k}<0$ and $\frac{\partial R}{\partial p}<0$. In addition, it is obvious that $\frac{\partial^2 C(n,n)}{\partial k\partial x}>0$.

For the sign of $\frac{\partial^2 R}{\partial p\partial x}$, I have

$$\frac{\partial^2 R}{\partial x\partial k}=\frac{[\frac{\partial^2 C(n,n)}{\partial x\partial k}y^\gamma\gamma-\frac{1}{\lambda}\frac{\partial R}{\partial k}\frac{\partial V}{\partial x}-\frac{R}{\lambda}\frac{\partial^2 V}{\partial x\partial k}]V-(\frac{\partial C(n,n)}{\partial x}y^\gamma\gamma-\frac{R}{\lambda}\frac{\partial V}{\partial x})\frac{\partial V}{\partial k}}{V^2}\lambda.$$

In addition,

$$\frac{\partial C(n,n)}{\partial x}=\frac{-(y^*)^{-\gamma}}{k[(\beta-\gamma)(r+\pi)-k\gamma]}[(\beta-\gamma)(\beta-1)(r+\pi)+k\gamma]<0;$$

therefore, $\frac{\partial V}{\partial x}<0$. It is known that $\frac{\partial^2 V}{\partial x\partial k}>0$ and $\frac{\partial V}{\partial k}>0$; hence, $\frac{\partial^2 R}{\partial x\partial k}<0$ and $\frac{\partial^2 R}{\partial p\partial x}<0$.

If $y^*_{FC}(n)<y^*_{CB}(n)$, the value and the risk premium in the continuation region are independent of $p(n)$. Therefore, $\frac{\partial R}{\partial p}=0$. ■

Proof of Proposition 2. When $n=N-1$ and $y^*_{FC}(n)>y^*_{CB}(n)$, in the continuation region,

$$\frac{\partial R}{\partial x}=\frac{\frac{\partial C(n,n)}{\partial x}y^\gamma\gamma-\frac{R}{\lambda}\frac{\partial V}{\partial x}}{V}\lambda>0.$$

In addition, the previous proposition proved $\frac{\partial^2 R}{\partial x\partial p}<0$. ■

Appendix B: Financial Constraints Indices

Lamont, Polk, and Saá-Requejo (2001) use the regression coefficients from Kaplan and Zingales (1997) to compute the KZ index as follows:

$$KZ = -1.001909 * CashFlow/K + 0.2826389 * Tobin's\ Q + 3.139193 * Debt/TotalCapital$$
$$-39.3678 * Dividends/K - 1.314759 * Cash/K,$$

where $CashFlow/K$ is computed as (Item 18 + Item14)/Item 8, $Tobin's\ Q$ as (Item 6 + CRSP December Market Equity - Item 60 - Item 74)/Item 6, $Debt/TotalCapital$ as (Item 9 + Item 34)/(Item 9 + Item 34 +Item 216), $Dividends/K$ as (Item 21 + Item 19)/Item 8, and $Cash/K$ as (Item 1/Item 8). Item numbers refer to Compustat annual data items as in the following: 1 (cash and short-term investments), 6 (liabilities and stockholders' equity–total), 8 (property, plant, and equipment), 9 (long-term debt–total), 14 (depreciation and amortization), 18 (income before extraordinary items), 19 (dividends–preferred), 21 (dividends–common), 34 (debt in current liabilities), 60 (common equity–total), 74 (deferred taxes), and 216 (stockholders' equity–total). Data item 8 is lagged. A firm needs to have valid information on all of the above annual items to be able to have a KZ index.

Following Hadlock and Pierce (2010), the SA index is calculated as

$$(-0.737 * Size) + (0.043 * Size^2) - (0.040 * Age),$$

where $Size$ equals the log of inflation-adjusted book assets, and Age is the number of years the firm is listed with a non-missing stock price on Compustat. In calculating this index, $Size$ is winsorized (i.e., capped) at (the log of) \$4.5 billion, and Age is winsorized at 37 years.

Following Whited and Wu (2006), the WW index is computed using Compustat quarterly data according to the following formula:

$$WW = -0.091 * CF - 0.062 * DIVPOS + 0.021 * TLTD - 0.044 * LNTA + 0.102 * ISG - 0.035 * SG,$$

where CF is the ratio of cash flow to total assets; $DIVPOS$ is an indicator that takes the value of one if the firm pays cash dividends; $TLTD$ is the ratio of the long-term debt to total assets; $LNTA$ is the natural log of total assets; ISG is the firm's three-digit industry sales growth; and SG is firm sales growth. All variables are deflated by the replacement cost of total assets as the sum of the replacement value of the capital stock plus the rest of the total assets. Whited (1992) details the computation of the replacement value of the capital stock.

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